Exchange Rate Dynamics and International Transmission in a Two-Country DSGE Model

Chiara Benedetti Kwame Asante Linh Nguyen

March 16, 2026

Abstract

We develop and estimate a two-country dynamic stochastic general equilibrium model with nominal rigidities, local currency pricing, and incomplete international financial markets. The model is calibrated to G7 data for 1973-2019. We find that the model can account for the high volatility and low cross-correlations of real exchange rates – the exchange rate disconnect puzzle – when combined with local currency pricing and incomplete markets. Variance decomposition reveals that monetary policy shocks account for 27 percent of real exchange rate variance, while productivity shocks explain 21 percent. Impulse responses show that a home productivity shock appreciates the real exchange rate on impact and improves the current account, with significant spillovers to the foreign country. A 25 basis point monetary tightening generates a real appreciation of 0.84 percent and contracts foreign output by 0.07 percent. These results are robust to alternative parameterizations of the trade elasticity, risk aversion, and degree of price stickiness.

Contents

I Introduction

This paper investigates the causal relationship between the treatment variable and the outcome variable. Understanding this relationship is critical for both economic theory and policy. Despite a growing literature, credible identification of the causal effect remains challenging due to potential confounders and reverse causality. We contribute to this literature by proposing a new identification strategy based on an instrumental variables strategy. Our approach addresses the key threats to internal validity that have limited the conclusions of prior work.

We contribute to the literature on international macro by exploiting a novel source of variation in the treatment variable. Our identification strategy relies on an instrumental variables strategy, which allows us to isolate the causal effect of the treatment variable on the outcome variable from confounding factors that have plagued earlier studies. The identifying assumption underlying our approach is that the proposed instrument affects the outcome variable only through its effect on the treatment variable, conditional on observable characteristics. We provide several pieces of evidence in support of this assumption, including placebo tests, overidentification tests, and bounds analysis.

The existing literature on the relationship between the treatment variable and the outcome variable has produced mixed results. Some studies find a positive and significant association, while others find no statistically significant effect. We argue that these discrepancies arise from differences in identification strategies and sample composition. Studies relying on cross-sectional variation tend to overstate the relationship due to omitted variable bias, while panel data approaches that rely on within-unit variation may attenuate the true effect by absorbing long-run changes in the treatment variable. Our approach navigates between these extremes by exploiting quasi-experimental variation.

A central question in international macro is whether changes in the treatment variable lead to changes in the outcome variable, or whether the observed correlation reflects reverse causality or omitted variable bias. We address this question using an instrumental

variables strategy applied to data from administrative records. The institutional setting we study provides a compelling natural experiment because the relevant policy changes were driven by factors orthogonal to contemporaneous economic conditions. We document this independence using a battery of balance tests and falsification exercises.

Our main finding is that the treatment variable has a statistically significant and economically meaningful effect on the outcome variable. The magnitude of this effect is robust across a wide range of specifications, subsamples, and alternative measures of the key variables. A one standard deviation increase in the treatment variable is associated with a change of approximately 0.3 to 0.5 standard deviations in the outcome variable, depending on the specification. This magnitude is consistent with the predictions of the theoretical model developed in Section 2 and is comparable to estimates from related quasi-experimental studies.

The remainder of this paper proceeds as follows. Section 2 presents the theoretical framework that motivates our empirical analysis and generates testable predictions. Section 3 describes the institutional background and the natural experiment we exploit. Section 4 describes the data sources and sample construction. Section 5 outlines our empirical strategy, including the identification assumptions and potential threats to validity. Section 6 presents the main results. Section 7 discusses robustness checks and extensions. Section 8 concludes with a discussion of policy implications.

Our paper relates to several strands of the economics literature. First, we contribute to the extensive body of work on the determinants of the outcome variable, which has been a central topic in international macro since the seminal contributions of the 1990s and 2000s. Second, we contribute to the methodological literature on an instrumental variables strategy by demonstrating its applicability to a new empirical context. Third, our findings speak to an ongoing policy debate about the effectiveness of interventions targeting the treatment variable.

The economic significance of our findings warrants emphasis. Back-of-the-envelope cal-

culations suggest that the estimated effect implies substantial welfare gains from policies that improve the treatment variable. Using the framework of Chetty (2009) for translating reduced-form estimates into welfare statements, we estimate that a policy intervention that increases the treatment variable by one standard deviation would generate welfare gains equivalent to approximately 2-4 percent of annual income. These calculations are necessarily approximate but suggest that the magnitudes involved are economically meaningful.

A distinctive feature of our analysis is the richness of the data. Unlike many previous studies that rely on aggregate proxies, our dataset contains individual-level information on both the outcome variable and the treatment variable, measured with high frequency over an extended time period. This granularity allows us to investigate heterogeneity in treatment effects across individuals, to construct precise event study estimates, and to conduct a range of placebo and falsification tests that would not be feasible with more aggregated data.

Before proceeding, we note several contributions relative to the closest existing work. While prior studies have documented a correlation between the treatment variable and the outcome variable in various contexts, our paper is among the first to provide credibly causal estimates using an instrumental variables strategy. Moreover, our analysis of mechanisms goes beyond what has been possible in earlier studies, shedding light on the channels through which the treatment variable affects the outcome variable.

This paper investigates the causal relationship between the treatment variable and the outcome variable. Understanding this relationship is critical for both economic theory and policy. Despite a growing literature, credible identification of the causal effect remains challenging due to potential confounders and reverse causality. We contribute to this literature by proposing a new identification strategy based on an instrumental variables strategy. Our approach addresses the key threats to internal validity that have limited the conclusions of prior work.

We contribute to the literature on international macro by exploiting a novel source of variation in the treatment variable. Our identification strategy relies on an instrumental variables strategy, which allows us to isolate the causal effect of the treatment variable on the outcome variable from confounding factors that have plagued earlier studies. The identifying assumption underlying our approach is that the proposed instrument affects the outcome variable only through its effect on the treatment variable, conditional on observable characteristics. We provide several pieces of evidence in support of this assumption, including placebo tests, overidentification tests, and bounds analysis.

The existing literature on the relationship between the treatment variable and the outcome variable has produced mixed results. Some studies find a positive and significant association, while others find no statistically significant effect. We argue that these discrepancies arise from differences in identification strategies and sample composition. Studies relying on cross-sectional variation tend to overstate the relationship due to omitted variable bias, while panel data approaches that rely on within-unit variation may attenuate the true effect by absorbing long-run changes in the treatment variable. Our approach navigates between these extremes by exploiting quasi-experimental variation.

A central question in international macro is whether changes in the treatment variable lead to changes in the outcome variable, or whether the observed correlation reflects reverse causality or omitted variable bias. We address this question using an instrumental variables strategy applied to data from administrative records. The institutional setting we study provides a compelling natural experiment because the relevant policy changes were driven by factors orthogonal to contemporaneous economic conditions. We document this independence using a battery of balance tests and falsification exercises.

Our main finding is that the treatment variable has a statistically significant and economically meaningful effect on the outcome variable. The magnitude of this effect is robust across a wide range of specifications, subsamples, and alternative measures of the key variables. A one standard deviation increase in the treatment variable is associated

with a change of approximately 0.3 to 0.5 standard deviations in the outcome variable, depending on the specification. This magnitude is consistent with the predictions of the theoretical model developed in Section 2 and is comparable to estimates from related quasi-experimental studies.

The remainder of this paper proceeds as follows. Section 2 presents the theoretical framework that motivates our empirical analysis and generates testable predictions. Section 3 describes the institutional background and the natural experiment we exploit. Section 4 describes the data sources and sample construction. Section 5 outlines our empirical strategy, including the identification assumptions and potential threats to validity. Section 6 presents the main results. Section 7 discusses robustness checks and extensions. Section 8 concludes with a discussion of policy implications.

Our paper relates to several strands of the economics literature. First, we contribute to the extensive body of work on the determinants of the outcome variable, which has been a central topic in international macro since the seminal contributions of the 1990s and 2000s. Second, we contribute to the methodological literature on an instrumental variables strategy by demonstrating its applicability to a new empirical context. Third, our findings speak to an ongoing policy debate about the effectiveness of interventions targeting the treatment variable.

The economic significance of our findings warrants emphasis. Back-of-the-envelope calculations suggest that the estimated effect implies substantial welfare gains from policies that improve the treatment variable. Using the framework of Chetty (2009) for translating reduced-form estimates into welfare statements, we estimate that a policy intervention that increases the treatment variable by one standard deviation would generate welfare gains equivalent to approximately 2-4 percent of annual income. These calculations are necessarily approximate but suggest that the magnitudes involved are economically meaningful.

A distinctive feature of our analysis is the richness of the data. Unlike many previous

studies that rely on aggregate proxies, our dataset contains individual-level information on both the outcome variable and the treatment variable, measured with high frequency over an extended time period. This granularity allows us to investigate heterogeneity in treatment effects across individuals, to construct precise event study estimates, and to conduct a range of placebo and falsification tests that would not be feasible with more aggregated data.

Before proceeding, we note several contributions relative to the closest existing work. While prior studies have documented a correlation between the treatment variable and the outcome variable in various contexts, our paper is among the first to provide credibly causal estimates using an instrumental variables strategy. Moreover, our analysis of mechanisms goes beyond what has been possible in earlier studies, shedding light on the channels through which the treatment variable affects the outcome variable.

II A Two-Country DSGE Model

Consider an economy populated by a continuum of agents indexed by $i \in [0, 1]$. Each agent chooses the outcome variable to maximize lifetime utility subject to a budget constraint. The key source of heterogeneity across agents is their exposure to the treatment variable. We solve for the equilibrium of this economy and derive comparative statics with respect to the key parameters. The model generates a clear testable prediction: an exogenous increase in the treatment variable should lead to a positive and significant change in the outcome variable, with the magnitude depending on structural parameters that we estimate in the empirical analysis.

Let Y_i denote the outcome of interest for unit i. We model Y_i as a function of treatment status D_i, observed covariates X_i, and unobserved heterogeneity ε_i. Under standard regularity conditions, the conditional expectation function takes the form presented in Equation **??**. The parameter of interest is the coefficient on treatment status, which captures

the causal effect of the treatment variable on the outcome variable holding constant observed and unobserved confounders. Identification of this parameter requires an exclusion restriction, which we discuss in detail below.

The theoretical framework builds on the canonical model of field. Agents face a discrete choice between J alternatives, each yielding utility $U_{ij} = V_{ij} + \epsilon_{ij}$, where V_{ij} is the deterministic component and ϵ_{ij} captures idiosyncratic preferences. Under standard distributional assumptions on the error term, the choice probabilities take a tractable closed form. The model can be estimated using maximum likelihood or method of moments, and the resulting structural parameters can be used to simulate counterfactual policy scenarios.

We derive the key prediction of the model through a simple comparative statics exercise. An exogenous increase in the treatment variable shifts the equilibrium allocation of the outcome variable through two channels: a direct effect operating through prices, and an indirect effect operating through the reallocation of factors across sectors. The total effect is ambiguous in general, but under the empirically relevant parameter values, we show that the direct effect dominates. Proposition 1 in Appendix A formalizes this result and establishes conditions under which the sign of the total effect is unambiguous.

Under the assumption that the proposed instrument satisfies the exclusion restriction, the model implies that the reduced-form relationship between the proposed instrument and the outcome variable is proportional to the structural parameter of interest. This motivates our instrumental variables strategy described in Section **??**. A key advantage of the IV approach is that it does not require functional form assumptions on the relationship between the treatment variable and the unobserved confounders. The cost is that we identify a local average treatment effect rather than the population average treatment effect.

The model delivers additional testable implications beyond the main prediction. First, the treatment effect should be larger for individuals with lower baseline levels of the outcome variable, reflecting diminishing marginal returns. Second, the effect should be persis-

tent over time rather than transitory, because the mechanism operates through permanent changes in the level of the treatment variable. Third, the effect should be concentrated in sectors with greater exposure to the underlying channel. We test each of these auxiliary predictions in Section 6.

We make several simplifying assumptions to maintain tractability. First, we assume that agents have rational expectations and perfect information about the distribution of shocks. Second, we abstract from general equilibrium effects by treating prices as exogenous to individual agents. Third, we assume that the production technology exhibits constant returns to scale. We discuss the robustness of our theoretical predictions to relaxing each of these assumptions in Appendix A, where we also present a more general version of the model.

The equilibrium of the model can be characterized by a system of first-order conditions and market-clearing conditions. In the interior solution, the marginal benefit of increasing the treatment variable equals the marginal cost, yielding an implicit equation for the optimal level of the treatment variable as a function of exogenous parameters and state variables. We log-linearize this equation around the steady state to obtain the estimating equation that motivates our empirical specification.

Consider an economy populated by a continuum of agents indexed by $i \in [0, 1]$. Each agent chooses the outcome variable to maximize lifetime utility subject to a budget constraint. The key source of heterogeneity across agents is their exposure to the treatment variable. We solve for the equilibrium of this economy and derive comparative statics with respect to the key parameters. The model generates a clear testable prediction: an exogenous increase in the treatment variable should lead to a positive and significant change in the outcome variable, with the magnitude depending on structural parameters that we estimate in the empirical analysis.

Let Y_i denote the outcome of interest for unit i. We model Y_i as a function of treatment status D_i, observed covariates X_i, and unobserved heterogeneity ε_i. Under standard reg-

ularity conditions, the conditional expectation function takes the form presented in Equation **??**. The parameter of interest is the coefficient on treatment status, which captures the causal effect of the treatment variable on the outcome variable holding constant observed and unobserved confounders. Identification of this parameter requires an exclusion restriction, which we discuss in detail below.

The theoretical framework builds on the canonical model of field. Agents face a discrete choice between J alternatives, each yielding utility $U_{ij} = V_{ij} + \epsilon_{ij}$, where V_{ij} is the deterministic component and ϵ_{ij} captures idiosyncratic preferences. Under standard distributional assumptions on the error term, the choice probabilities take a tractable closed form. The model can be estimated using maximum likelihood or method of moments, and the resulting structural parameters can be used to simulate counterfactual policy scenarios.

We derive the key prediction of the model through a simple comparative statics exercise. An exogenous increase in the treatment variable shifts the equilibrium allocation of the outcome variable through two channels: a direct effect operating through prices, and an indirect effect operating through the reallocation of factors across sectors. The total effect is ambiguous in general, but under the empirically relevant parameter values, we show that the direct effect dominates. Proposition 1 in Appendix A formalizes this result and establishes conditions under which the sign of the total effect is unambiguous.

Under the assumption that the proposed instrument satisfies the exclusion restriction, the model implies that the reduced-form relationship between the proposed instrument and the outcome variable is proportional to the structural parameter of interest. This motivates our instrumental variables strategy described in Section **??**. A key advantage of the IV approach is that it does not require functional form assumptions on the relationship between the treatment variable and the unobserved confounders. The cost is that we identify a local average treatment effect rather than the population average treatment effect.

The model delivers additional testable implications beyond the main prediction. First,

the treatment effect should be larger for individuals with lower baseline levels of the outcome variable, reflecting diminishing marginal returns. Second, the effect should be persistent over time rather than transitory, because the mechanism operates through permanent changes in the level of the treatment variable. Third, the effect should be concentrated in sectors with greater exposure to the underlying channel. We test each of these auxiliary predictions in Section 6.

We make several simplifying assumptions to maintain tractability. First, we assume that agents have rational expectations and perfect information about the distribution of shocks. Second, we abstract from general equilibrium effects by treating prices as exogenous to individual agents. Third, we assume that the production technology exhibits constant returns to scale. We discuss the robustness of our theoretical predictions to relaxing each of these assumptions in Appendix A, where we also present a more general version of the model.

The equilibrium of the model can be characterized by a system of first-order conditions and market-clearing conditions. In the interior solution, the marginal benefit of increasing the treatment variable equals the marginal cost, yielding an implicit equation for the optimal level of the treatment variable as a function of exogenous parameters and state variables. We log-linearize this equation around the steady state to obtain the estimating equation that motivates our empirical specification.

Consider an economy populated by a continuum of agents indexed by $i \in [0, 1]$. Each agent chooses the outcome variable to maximize lifetime utility subject to a budget constraint. The key source of heterogeneity across agents is their exposure to the treatment variable. We solve for the equilibrium of this economy and derive comparative statics with respect to the key parameters. The model generates a clear testable prediction: an exogenous increase in the treatment variable should lead to a positive and significant change in the outcome variable, with the magnitude depending on structural parameters that we estimate in the empirical analysis.

Let Y_i denote the outcome of interest for unit i. We model Y_i as a function of treatment status D_i, observed covariates X_i, and unobserved heterogeneity ε_i. Under standard regularity conditions, the conditional expectation function takes the form presented in Equation **??**. The parameter of interest is the coefficient on treatment status, which captures the causal effect of the treatment variable on the outcome variable holding constant observed and unobserved confounders. Identification of this parameter requires an exclusion restriction, which we discuss in detail below.

A Households

This paper investigates the causal relationship between the treatment variable and the outcome variable. Understanding this relationship is critical for both economic theory and policy. Despite a growing literature, credible identification of the causal effect remains challenging due to potential confounders and reverse causality. We contribute to this literature by proposing a new identification strategy based on an instrumental variables strategy. Our approach addresses the key threats to internal validity that have limited the conclusions of prior work.

We contribute to the literature on international macro by exploiting a novel source of variation in the treatment variable. Our identification strategy relies on an instrumental variables strategy, which allows us to isolate the causal effect of the treatment variable on the outcome variable from confounding factors that have plagued earlier studies. The identifying assumption underlying our approach is that the proposed instrument affects the outcome variable only through its effect on the treatment variable, conditional on observable characteristics. We provide several pieces of evidence in support of this assumption, including placebo tests, overidentification tests, and bounds analysis.

The existing literature on the relationship between the treatment variable and the outcome variable has produced mixed results. Some studies find a positive and significant association, while others find no statistically significant effect. We argue that these discrep-

ancies arise from differences in identification strategies and sample composition. Studies relying on cross-sectional variation tend to overstate the relationship due to omitted variable bias, while panel data approaches that rely on within-unit variation may attenuate the true effect by absorbing long-run changes in the treatment variable. Our approach navigates between these extremes by exploiting quasi-experimental variation.

A central question in international macro is whether changes in the treatment variable lead to changes in the outcome variable, or whether the observed correlation reflects reverse causality or omitted variable bias. We address this question using an instrumental variables strategy applied to data from administrative records. The institutional setting we study provides a compelling natural experiment because the relevant policy changes were driven by factors orthogonal to contemporaneous economic conditions. We document this independence using a battery of balance tests and falsification exercises.

Our main finding is that the treatment variable has a statistically significant and economically meaningful effect on the outcome variable. The magnitude of this effect is robust across a wide range of specifications, subsamples, and alternative measures of the key variables. A one standard deviation increase in the treatment variable is associated with a change of approximately 0.3 to 0.5 standard deviations in the outcome variable, depending on the specification. This magnitude is consistent with the predictions of the theoretical model developed in Section 2 and is comparable to estimates from related quasi-experimental studies.

$$E_t\left[\beta\left(\frac{\dot{C}_{t+1}}{\dot{C}_t}\right)^{-\sigma}\frac{P_t}{P_{t+1}}(1+i_t)\right]=1,\quad \ddot{C}_t\equiv\frac{d^2C}{dt^2},\quad \mathcal{F}_t=\sigma\big\{C_s,P_s:s\leq t\big\} \tag{1}$$

$$P_tC_t+B_t+S_tB_t^*=(1+i_{t-1})B_{t-1}+S_t(1+i_{t-1}^*)B_{t-1}^*+W_tL_t+T_t \tag{2}$$

$$C_t=\left[(1-\alpha)^{1/\theta}C_{H,t}^{(\theta-1)/\theta}+\alpha^{1/\theta}C_{F,t}^{(\theta-1)/\theta}\right]^{\theta/(\theta-1)} \tag{3}$$

The remainder of this paper proceeds as follows. Section 2 presents the theoretical framework that motivates our empirical analysis and generates testable predictions. Section 3 describes the institutional background and the natural experiment we exploit. Section 4 describes the data sources and sample construction. Section 5 outlines our empirical strategy, including the identification assumptions and potential threats to validity. Section 6 presents the main results. Section 7 discusses robustness checks and extensions. Section 8 concludes with a discussion of policy implications.

Our paper relates to several strands of the economics literature. First, we contribute to the extensive body of work on the determinants of the outcome variable, which has been a central topic in international macro since the seminal contributions of the 1990s and 2000s. Second, we contribute to the methodological literature on an instrumental variables strategy by demonstrating its applicability to a new empirical context. Third, our findings speak to an ongoing policy debate about the effectiveness of interventions targeting the treatment variable.

The economic significance of our findings warrants emphasis. Back-of-the-envelope calculations suggest that the estimated effect implies substantial welfare gains from policies that improve the treatment variable. Using the framework of Chetty (2009) for translating reduced-form estimates into welfare statements, we estimate that a policy intervention that increases the treatment variable by one standard deviation would generate welfare gains equivalent to approximately 2-4 percent of annual income. These calculations are necessarily approximate but suggest that the magnitudes involved are economically meaningful.

A distinctive feature of our analysis is the richness of the data. Unlike many previous studies that rely on aggregate proxies, our dataset contains individual-level information on both the outcome variable and the treatment variable, measured with high frequency over an extended time period. This granularity allows us to investigate heterogeneity in treatment effects across individuals, to construct precise event study estimates, and to

conduct a range of placebo and falsification tests that would not be feasible with more aggregated data.

Before proceeding, we note several contributions relative to the closest existing work. While prior studies have documented a correlation between the treatment variable and the outcome variable in various contexts, our paper is among the first to provide credibly causal estimates using an instrumental variables strategy. Moreover, our analysis of mechanisms goes beyond what has been possible in earlier studies, shedding light on the channels through which the treatment variable affects the outcome variable.

This paper investigates the causal relationship between the treatment variable and the outcome variable. Understanding this relationship is critical for both economic theory and policy. Despite a growing literature, credible identification of the causal effect remains challenging due to potential confounders and reverse causality. We contribute to this literature by proposing a new identification strategy based on an instrumental variables strategy. Our approach addresses the key threats to internal validity that have limited the conclusions of prior work.

We contribute to the literature on international macro by exploiting a novel source of variation in the treatment variable. Our identification strategy relies on an instrumental variables strategy, which allows us to isolate the causal effect of the treatment variable on the outcome variable from confounding factors that have plagued earlier studies. The identifying assumption underlying our approach is that the proposed instrument affects the outcome variable only through its effect on the treatment variable, conditional on observable characteristics. We provide several pieces of evidence in support of this assumption, including placebo tests, overidentification tests, and bounds analysis.

The existing literature on the relationship between the treatment variable and the outcome variable has produced mixed results. Some studies find a positive and significant association, while others find no statistically significant effect. We argue that these discrepancies arise from differences in identification strategies and sample composition. Studies

relying on cross-sectional variation tend to overstate the relationship due to omitted variable bias, while panel data approaches that rely on within-unit variation may attenuate the true effect by absorbing long-run changes in the treatment variable. Our approach navigates between these extremes by exploiting quasi-experimental variation.

B Firms and Price Setting

This paper investigates the causal relationship between the treatment variable and the outcome variable. Understanding this relationship is critical for both economic theory and policy. Despite a growing literature, credible identification of the causal effect remains challenging due to potential confounders and reverse causality. We contribute to this literature by proposing a new identification strategy based on an instrumental variables strategy. Our approach addresses the key threats to internal validity that have limited the conclusions of prior work.

We contribute to the literature on international macro by exploiting a novel source of variation in the treatment variable. Our identification strategy relies on an instrumental variables strategy, which allows us to isolate the causal effect of the treatment variable on the outcome variable from confounding factors that have plagued earlier studies. The identifying assumption underlying our approach is that the proposed instrument affects the outcome variable only through its effect on the treatment variable, conditional on observable characteristics. We provide several pieces of evidence in support of this assumption, including placebo tests, overidentification tests, and bounds analysis.

The existing literature on the relationship between the treatment variable and the outcome variable has produced mixed results. Some studies find a positive and significant association, while others find no statistically significant effect. We argue that these discrepancies arise from differences in identification strategies and sample composition. Studies relying on cross-sectional variation tend to overstate the relationship due to omitted variable bias, while panel data approaches that rely on within-unit variation may attenuate

the true effect by absorbing long-run changes in the treatment variable. Our approach navigates between these extremes by exploiting quasi-experimental variation.

A central question in international macro is whether changes in the treatment variable lead to changes in the outcome variable, or whether the observed correlation reflects reverse causality or omitted variable bias. We address this question using an instrumental variables strategy applied to data from administrative records. The institutional setting we study provides a compelling natural experiment because the relevant policy changes were driven by factors orthogonal to contemporaneous economic conditions. We document this independence using a battery of balance tests and falsification exercises.

$$\pi_{H,t} = \beta E_t \pi_{H,t+1} + \kappa \hat{mc}_t, \quad \kappa = \frac{(1-\xi)(1-\beta\xi)}{\xi}, \quad \dot{p}_{H,t} \equiv \iint_{\mathcal{F}_t} \frac{\partial^2 p_H}{\partial z \partial \omega} \, dz \, d\omega \tag{4}$$

Our main finding is that the treatment variable has a statistically significant and economically meaningful effect on the outcome variable. The magnitude of this effect is robust across a wide range of specifications, subsamples, and alternative measures of the key variables. A one standard deviation increase in the treatment variable is associated with a change of approximately 0.3 to 0.5 standard deviations in the outcome variable, depending on the specification. This magnitude is consistent with the predictions of the theoretical model developed in Section 2 and is comparable to estimates from related quasi-experimental studies.

The remainder of this paper proceeds as follows. Section 2 presents the theoretical framework that motivates our empirical analysis and generates testable predictions. Section 3 describes the institutional background and the natural experiment we exploit. Section 4 describes the data sources and sample construction. Section 5 outlines our empirical strategy, including the identification assumptions and potential threats to validity. Section 6 presents the main results. Section 7 discusses robustness checks and extensions. Section 8 concludes with a discussion of policy implications.

Our paper relates to several strands of the economics literature. First, we contribute

to the extensive body of work on the determinants of the outcome variable, which has been a central topic in international macro since the seminal contributions of the 1990s and 2000s. Second, we contribute to the methodological literature on an instrumental variables strategy by demonstrating its applicability to a new empirical context. Third, our findings speak to an ongoing policy debate about the effectiveness of interventions targeting the treatment variable.

The economic significance of our findings warrants emphasis. Back-of-the-envelope calculations suggest that the estimated effect implies substantial welfare gains from policies that improve the treatment variable. Using the framework of Chetty (2009) for translating reduced-form estimates into welfare statements, we estimate that a policy intervention that increases the treatment variable by one standard deviation would generate welfare gains equivalent to approximately 2-4 percent of annual income. These calculations are necessarily approximate but suggest that the magnitudes involved are economically meaningful.

A distinctive feature of our analysis is the richness of the data. Unlike many previous studies that rely on aggregate proxies, our dataset contains individual-level information on both the outcome variable and the treatment variable, measured with high frequency over an extended time period. This granularity allows us to investigate heterogeneity in treatment effects across individuals, to construct precise event study estimates, and to conduct a range of placebo and falsification tests that would not be feasible with more aggregated data.

Before proceeding, we note several contributions relative to the closest existing work. While prior studies have documented a correlation between the treatment variable and the outcome variable in various contexts, our paper is among the first to provide credibly causal estimates using an instrumental variables strategy. Moreover, our analysis of mechanisms goes beyond what has been possible in earlier studies, shedding light on the channels through which the treatment variable affects the outcome variable.

C Government and Monetary Policy

This paper investigates the causal relationship between the treatment variable and the outcome variable. Understanding this relationship is critical for both economic theory and policy. Despite a growing literature, credible identification of the causal effect remains challenging due to potential confounders and reverse causality. We contribute to this literature by proposing a new identification strategy based on an instrumental variables strategy. Our approach addresses the key threats to internal validity that have limited the conclusions of prior work.

We contribute to the literature on international macro by exploiting a novel source of variation in the treatment variable. Our identification strategy relies on an instrumental variables strategy, which allows us to isolate the causal effect of the treatment variable on the outcome variable from confounding factors that have plagued earlier studies. The identifying assumption underlying our approach is that the proposed instrument affects the outcome variable only through its effect on the treatment variable, conditional on observable characteristics. We provide several pieces of evidence in support of this assumption, including placebo tests, overidentification tests, and bounds analysis.

The existing literature on the relationship between the treatment variable and the outcome variable has produced mixed results. Some studies find a positive and significant association, while others find no statistically significant effect. We argue that these discrepancies arise from differences in identification strategies and sample composition. Studies relying on cross-sectional variation tend to overstate the relationship due to omitted variable bias, while panel data approaches that rely on within-unit variation may attenuate the true effect by absorbing long-run changes in the treatment variable. Our approach navigates between these extremes by exploiting quasi-experimental variation.

$$i_t = \rho_i i_{t-1} + (1 - \rho_i) \left[\bar{\imath} + \phi_\pi \pi_t + \phi_y \hat{y}_t \right] + \varepsilon_t^m \tag{5}$$

A central question in international macro is whether changes in the treatment variable lead to changes in the outcome variable, or whether the observed correlation reflects reverse causality or omitted variable bias. We address this question using an instrumental variables strategy applied to data from administrative records. The institutional setting we study provides a compelling natural experiment because the relevant policy changes were driven by factors orthogonal to contemporaneous economic conditions. We document this independence using a battery of balance tests and falsification exercises.

Our main finding is that the treatment variable has a statistically significant and economically meaningful effect on the outcome variable. The magnitude of this effect is robust across a wide range of specifications, subsamples, and alternative measures of the key variables. A one standard deviation increase in the treatment variable is associated with a change of approximately 0.3 to 0.5 standard deviations in the outcome variable, depending on the specification. This magnitude is consistent with the predictions of the theoretical model developed in Section 2 and is comparable to estimates from related quasi-experimental studies.

The remainder of this paper proceeds as follows. Section 2 presents the theoretical framework that motivates our empirical analysis and generates testable predictions. Section 3 describes the institutional background and the natural experiment we exploit. Section 4 describes the data sources and sample construction. Section 5 outlines our empirical strategy, including the identification assumptions and potential threats to validity. Section 6 presents the main results. Section 7 discusses robustness checks and extensions. Section 8 concludes with a discussion of policy implications.

Our paper relates to several strands of the economics literature. First, we contribute to the extensive body of work on the determinants of the outcome variable, which has been a central topic in international macro since the seminal contributions of the 1990s and 2000s. Second, we contribute to the methodological literature on an instrumental variables strategy by demonstrating its applicability to a new empirical context. Third,

our findings speak to an ongoing policy debate about the effectiveness of interventions targeting the treatment variable.

The economic significance of our findings warrants emphasis. Back-of-the-envelope calculations suggest that the estimated effect implies substantial welfare gains from policies that improve the treatment variable. Using the framework of Chetty (2009) for translating reduced-form estimates into welfare statements, we estimate that a policy intervention that increases the treatment variable by one standard deviation would generate welfare gains equivalent to approximately 2-4 percent of annual income. These calculations are necessarily approximate but suggest that the magnitudes involved are economically meaningful.

D Equilibrium and Market Clearing

This paper investigates the causal relationship between the treatment variable and the outcome variable. Understanding this relationship is critical for both economic theory and policy. Despite a growing literature, credible identification of the causal effect remains challenging due to potential confounders and reverse causality. We contribute to this literature by proposing a new identification strategy based on an instrumental variables strategy. Our approach addresses the key threats to internal validity that have limited the conclusions of prior work.

We contribute to the literature on international macro by exploiting a novel source of variation in the treatment variable. Our identification strategy relies on an instrumental variables strategy, which allows us to isolate the causal effect of the treatment variable on the outcome variable from confounding factors that have plagued earlier studies. The identifying assumption underlying our approach is that the proposed instrument affects the outcome variable only through its effect on the treatment variable, conditional on observable characteristics. We provide several pieces of evidence in support of this assumption, including placebo tests, overidentification tests, and bounds analysis.

The existing literature on the relationship between the treatment variable and the outcome variable has produced mixed results. Some studies find a positive and significant association, while others find no statistically significant effect. We argue that these discrepancies arise from differences in identification strategies and sample composition. Studies relying on cross-sectional variation tend to overstate the relationship due to omitted variable bias, while panel data approaches that rely on within-unit variation may attenuate the true effect by absorbing long-run changes in the treatment variable. Our approach navigates between these extremes by exploiting quasi-experimental variation.

$$i_t - i_t^* = E_t \Delta s_{t+1} + \phi_b b_t + \xi_t^{UIP} \tag{6}$$

$$\text{CA}_t = \frac{S_t B_t^* - S_{t-1} B_{t-1}^*}{P_t Y_t} = \text{NX}_t + (i_{t-1}^* - \pi_t^*) \frac{S_t B_{t-1}^*}{P_t Y_t} \tag{7}$$

$$T_t = \frac{P_{F,t}}{P_{H,t}} = \frac{S_t P_{F,t}^*}{P_{H,t}} \tag{8}$$

$$Y_t = C_{H,t} + C_{H,t}^* + G_t, \quad C_{H,t}^* = \alpha^* \left(\frac{S_t P_{H,t}}{P_t^*} \right)^{-\theta} C_t^* \tag{9}$$

$$\dot{q}_t = \dot{s}_t + \dot{p}_t^* - \dot{p}_t, \quad q_t = (2\alpha - 1) T_t + (2\alpha^* - 1) T_t^*, \quad \ddot{q}_t = \mathcal{F}\left(\dot{T}_t, \dot{T}_t^* \right) \tag{10}$$

A central question in international macro is whether changes in the treatment variable lead to changes in the outcome variable, or whether the observed correlation reflects reverse causality or omitted variable bias. We address this question using an instrumental variables strategy applied to data from administrative records. The institutional setting we study provides a compelling natural experiment because the relevant policy changes were driven by factors orthogonal to contemporaneous economic conditions. We document this independence using a battery of balance tests and falsification exercises.

Our main finding is that the treatment variable has a statistically significant and economically meaningful effect on the outcome variable. The magnitude of this effect is robust across a wide range of specifications, subsamples, and alternative measures of the key variables. A one standard deviation increase in the treatment variable is associated with a change of approximately 0.3 to 0.5 standard deviations in the outcome variable, depending on the specification. This magnitude is consistent with the predictions of the theoretical model developed in Section 2 and is comparable to estimates from related quasi-experimental studies.

The remainder of this paper proceeds as follows. Section 2 presents the theoretical framework that motivates our empirical analysis and generates testable predictions. Section 3 describes the institutional background and the natural experiment we exploit. Section 4 describes the data sources and sample construction. Section 5 outlines our empirical strategy, including the identification assumptions and potential threats to validity. Section 6 presents the main results. Section 7 discusses robustness checks and extensions. Section 8 concludes with a discussion of policy implications.

Our paper relates to several strands of the economics literature. First, we contribute to the extensive body of work on the determinants of the outcome variable, which has been a central topic in international macro since the seminal contributions of the 1990s and 2000s. Second, we contribute to the methodological literature on an instrumental variables strategy by demonstrating its applicability to a new empirical context. Third, our findings speak to an ongoing policy debate about the effectiveness of interventions targeting the treatment variable.

The economic significance of our findings warrants emphasis. Back-of-the-envelope calculations suggest that the estimated effect implies substantial welfare gains from policies that improve the treatment variable. Using the framework of Chetty (2009) for translating reduced-form estimates into welfare statements, we estimate that a policy intervention that increases the treatment variable by one standard deviation would generate welfare

gains equivalent to approximately 2-4 percent of annual income. These calculations are necessarily approximate but suggest that the magnitudes involved are economically meaningful.

III Calibration

This paper investigates the causal relationship between the treatment variable and the outcome variable. Understanding this relationship is critical for both economic theory and policy. Despite a growing literature, credible identification of the causal effect remains challenging due to potential confounders and reverse causality. We contribute to this literature by proposing a new identification strategy based on an instrumental variables strategy. Our approach addresses the key threats to internal validity that have limited the conclusions of prior work.

We contribute to the literature on international macro by exploiting a novel source of variation in the treatment variable. Our identification strategy relies on an instrumental variables strategy, which allows us to isolate the causal effect of the treatment variable on the outcome variable from confounding factors that have plagued earlier studies. The identifying assumption underlying our approach is that the proposed instrument affects the outcome variable only through its effect on the treatment variable, conditional on observable characteristics. We provide several pieces of evidence in support of this assumption, including placebo tests, overidentification tests, and bounds analysis.

The existing literature on the relationship between the treatment variable and the outcome variable has produced mixed results. Some studies find a positive and significant association, while others find no statistically significant effect. We argue that these discrepancies arise from differences in identification strategies and sample composition. Studies relying on cross-sectional variation tend to overstate the relationship due to omitted variable bias, while panel data approaches that rely on within-unit variation may attenuate

the true effect by absorbing long-run changes in the treatment variable. Our approach navigates between these extremes by exploiting quasi-experimental variation.

A central question in international macro is whether changes in the treatment variable lead to changes in the outcome variable, or whether the observed correlation reflects reverse causality or omitted variable bias. We address this question using an instrumental variables strategy applied to data from administrative records. The institutional setting we study provides a compelling natural experiment because the relevant policy changes were driven by factors orthogonal to contemporaneous economic conditions. We document this independence using a battery of balance tests and falsification exercises.

Our main finding is that the treatment variable has a statistically significant and economically meaningful effect on the outcome variable. The magnitude of this effect is robust across a wide range of specifications, subsamples, and alternative measures of the key variables. A one standard deviation increase in the treatment variable is associated with a change of approximately 0.3 to 0.5 standard deviations in the outcome variable, depending on the specification. This magnitude is consistent with the predictions of the theoretical model developed in Section 2 and is comparable to estimates from related quasi-experimental studies.

The remainder of this paper proceeds as follows. Section 2 presents the theoretical framework that motivates our empirical analysis and generates testable predictions. Section 3 describes the institutional background and the natural experiment we exploit. Section 4 describes the data sources and sample construction. Section 5 outlines our empirical strategy, including the identification assumptions and potential threats to validity. Section 6 presents the main results. Section 7 discusses robustness checks and extensions. Section 8 concludes with a discussion of policy implications.

Our paper relates to several strands of the economics literature. First, we contribute to the extensive body of work on the determinants of the outcome variable, which has been a central topic in international macro since the seminal contributions of the 1990s

and 2000s. Second, we contribute to the methodological literature on an instrumental variables strategy by demonstrating its applicability to a new empirical context. Third, our findings speak to an ongoing policy debate about the effectiveness of interventions targeting the treatment variable.

The economic significance of our findings warrants emphasis. Back-of-the-envelope calculations suggest that the estimated effect implies substantial welfare gains from policies that improve the treatment variable. Using the framework of Chetty (2009) for translating reduced-form estimates into welfare statements, we estimate that a policy intervention that increases the treatment variable by one standard deviation would generate welfare gains equivalent to approximately 2-4 percent of annual income. These calculations are necessarily approximate but suggest that the magnitudes involved are economically meaningful.

A distinctive feature of our analysis is the richness of the data. Unlike many previous studies that rely on aggregate proxies, our dataset contains individual-level information on both the outcome variable and the treatment variable, measured with high frequency over an extended time period. This granularity allows us to investigate heterogeneity in treatment effects across individuals, to construct precise event study estimates, and to conduct a range of placebo and falsification tests that would not be feasible with more aggregated data.

Before proceeding, we note several contributions relative to the closest existing work. While prior studies have documented a correlation between the treatment variable and the outcome variable in various contexts, our paper is among the first to provide credibly causal estimates using an instrumental variables strategy. Moreover, our analysis of mechanisms goes beyond what has been possible in earlier studies, shedding light on the channels through which the treatment variable affects the outcome variable.

This paper investigates the causal relationship between the treatment variable and the outcome variable. Understanding this relationship is critical for both economic theory and

Table 1: Calibration Parameters

	Parameter	Symbol	Value	Source
Discount factor	β	0.99	Standard	
Inverse Frisch elasticity	ϕ	1.00	Kimball (1995)	
Risk aversion	σ	2.00	Obstfeld-Rogoff (1996)	
Trade elasticity	θ	1.50	Hooper-Marquez (1995)	
Home bias	α	0.72	Trade data	
Calvo price-stickiness	ξ	0.75	Gali-Gertler (1999)	
Taylor rule weight on inflation	ϕ_π	1.50	Taylor (1993)	
Taylor rule weight on output	ϕ_y	0.50	Taylor (1993)	
Steady-state trade share	γ	0.28	BEA/Eurostat	
AR(1) coefficient productivity	ρ_a	0.90	Backus et al. (1992)	
Std. dev. productivity shock	σ_a	0.0071	Backus et al. (1992)	

policy. Despite a growing literature, credible identification of the causal effect remains challenging due to potential confounders and reverse causality. We contribute to this literature by proposing a new identification strategy based on an instrumental variables strategy. Our approach addresses the key threats to internal validity that have limited the conclusions of prior work.

We contribute to the literature on international macro by exploiting a novel source of variation in the treatment variable. Our identification strategy relies on an instrumental variables strategy, which allows us to isolate the causal effect of the treatment variable on the outcome variable from confounding factors that have plagued earlier studies. The identifying assumption underlying our approach is that the proposed instrument affects the outcome variable only through its effect on the treatment variable, conditional on observable characteristics. We provide several pieces of evidence in support of this assumption, including placebo tests, overidentification tests, and bounds analysis.

The existing literature on the relationship between the treatment variable and the outcome variable has produced mixed results. Some studies find a positive and significant association, while others find no statistically significant effect. We argue that these discrepancies arise from differences in identification strategies and sample composition. Studies

relying on cross-sectional variation tend to overstate the relationship due to omitted variable bias, while panel data approaches that rely on within-unit variation may attenuate the true effect by absorbing long-run changes in the treatment variable. Our approach navigates between these extremes by exploiting quasi-experimental variation.

A central question in international macro is whether changes in the treatment variable lead to changes in the outcome variable, or whether the observed correlation reflects reverse causality or omitted variable bias. We address this question using an instrumental variables strategy applied to data from administrative records. The institutional setting we study provides a compelling natural experiment because the relevant policy changes were driven by factors orthogonal to contemporaneous economic conditions. We document this independence using a battery of balance tests and falsification exercises.

A Parameter Values and Sources

This paper investigates the causal relationship between the treatment variable and the outcome variable. Understanding this relationship is critical for both economic theory and policy. Despite a growing literature, credible identification of the causal effect remains challenging due to potential confounders and reverse causality. We contribute to this literature by proposing a new identification strategy based on an instrumental variables strategy. Our approach addresses the key threats to internal validity that have limited the conclusions of prior work.

We contribute to the literature on international macro by exploiting a novel source of variation in the treatment variable. Our identification strategy relies on an instrumental variables strategy, which allows us to isolate the causal effect of the treatment variable on the outcome variable from confounding factors that have plagued earlier studies. The identifying assumption underlying our approach is that the proposed instrument affects the outcome variable only through its effect on the treatment variable, conditional on observable characteristics. We provide several pieces of evidence in support of this assumption,

including placebo tests, overidentification tests, and bounds analysis.

The existing literature on the relationship between the treatment variable and the outcome variable has produced mixed results. Some studies find a positive and significant association, while others find no statistically significant effect. We argue that these discrepancies arise from differences in identification strategies and sample composition. Studies relying on cross-sectional variation tend to overstate the relationship due to omitted variable bias, while panel data approaches that rely on within-unit variation may attenuate the true effect by absorbing long-run changes in the treatment variable. Our approach navigates between these extremes by exploiting quasi-experimental variation.

A central question in international macro is whether changes in the treatment variable lead to changes in the outcome variable, or whether the observed correlation reflects reverse causality or omitted variable bias. We address this question using an instrumental variables strategy applied to data from administrative records. The institutional setting we study provides a compelling natural experiment because the relevant policy changes were driven by factors orthogonal to contemporaneous economic conditions. We document this independence using a battery of balance tests and falsification exercises.

Our main finding is that the treatment variable has a statistically significant and economically meaningful effect on the outcome variable. The magnitude of this effect is robust across a wide range of specifications, subsamples, and alternative measures of the key variables. A one standard deviation increase in the treatment variable is associated with a change of approximately 0.3 to 0.5 standard deviations in the outcome variable, depending on the specification. This magnitude is consistent with the predictions of the theoretical model developed in Section 2 and is comparable to estimates from related quasi-experimental studies.

The remainder of this paper proceeds as follows. Section 2 presents the theoretical framework that motivates our empirical analysis and generates testable predictions. Section 3 describes the institutional background and the natural experiment we exploit. Sec-

tion 4 describes the data sources and sample construction. Section 5 outlines our empirical strategy, including the identification assumptions and potential threats to validity. Section 6 presents the main results. Section 7 discusses robustness checks and extensions. Section 8 concludes with a discussion of policy implications.

Our paper relates to several strands of the economics literature. First, we contribute to the extensive body of work on the determinants of the outcome variable, which has been a central topic in international macro since the seminal contributions of the 1990s and 2000s. Second, we contribute to the methodological literature on an instrumental variables strategy by demonstrating its applicability to a new empirical context. Third, our findings speak to an ongoing policy debate about the effectiveness of interventions targeting the treatment variable.

The economic significance of our findings warrants emphasis. Back-of-the-envelope calculations suggest that the estimated effect implies substantial welfare gains from policies that improve the treatment variable. Using the framework of Chetty (2009) for translating reduced-form estimates into welfare statements, we estimate that a policy intervention that increases the treatment variable by one standard deviation would generate welfare gains equivalent to approximately 2-4 percent of annual income. These calculations are necessarily approximate but suggest that the magnitudes involved are economically meaningful.

B Model Fit: Moments vs. Data

Consider an economy populated by a continuum of agents indexed by $i \in [0, 1]$. Each agent chooses the outcome variable to maximize lifetime utility subject to a budget constraint. The key source of heterogeneity across agents is their exposure to the treatment variable. We solve for the equilibrium of this economy and derive comparative statics with respect to the key parameters. The model generates a clear testable prediction: an exogenous increase in the treatment variable should lead to a positive and significant change

in the outcome variable, with the magnitude depending on structural parameters that we estimate in the empirical analysis.

Let Y_i denote the outcome of interest for unit i. We model Y_i as a function of treatment status D_i, observed covariates X_i, and unobserved heterogeneity ε_i. Under standard regularity conditions, the conditional expectation function takes the form presented in Equation **??**. The parameter of interest is the coefficient on treatment status, which captures the causal effect of the treatment variable on the outcome variable holding constant observed and unobserved confounders. Identification of this parameter requires an exclusion restriction, which we discuss in detail below.

The theoretical framework builds on the canonical model of field. Agents face a discrete choice between J alternatives, each yielding utility $U_{ij} = V_{ij} + \epsilon_{ij}$, where V_{ij} is the deterministic component and ϵ_{ij} captures idiosyncratic preferences. Under standard distributional assumptions on the error term, the choice probabilities take a tractable closed form. The model can be estimated using maximum likelihood or method of moments, and the resulting structural parameters can be used to simulate counterfactual policy scenarios.

We derive the key prediction of the model through a simple comparative statics exercise. An exogenous increase in the treatment variable shifts the equilibrium allocation of the outcome variable through two channels: a direct effect operating through prices, and an indirect effect operating through the reallocation of factors across sectors. The total effect is ambiguous in general, but under the empirically relevant parameter values, we show that the direct effect dominates. Proposition 1 in Appendix A formalizes this result and establishes conditions under which the sign of the total effect is unambiguous.

Under the assumption that the proposed instrument satisfies the exclusion restriction, the model implies that the reduced-form relationship between the proposed instrument and the outcome variable is proportional to the structural parameter of interest. This motivates our instrumental variables strategy described in Section **??**. A key advantage of the IV approach is that it does not require functional form assumptions on the relationship

between the treatment variable and the unobserved confounders. The cost is that we identify a local average treatment effect rather than the population average treatment effect.

The model delivers additional testable implications beyond the main prediction. First, the treatment effect should be larger for individuals with lower baseline levels of the outcome variable, reflecting diminishing marginal returns. Second, the effect should be persistent over time rather than transitory, because the mechanism operates through permanent changes in the level of the treatment variable. Third, the effect should be concentrated in sectors with greater exposure to the underlying channel. We test each of these auxiliary predictions in Section 6.

Table 2: Model Moments vs. Data: Business Cycle Statistics

	Data	Baseline	No Rigidities	Financial Frictions
Panel A: Volatilities (% std. dev.)				
Output	1.68	1.71	1.89	1.74
Consumption	1.27	1.24	1.41	1.31
Investment	5.61	5.43	6.12	5.87
Real exchange rate	11.24	9.87	4.32	10.14
Terms of trade	3.14	3.08	2.11	3.19
Current account / GDP	1.82	1.76	1.53	1.91
Panel B: Cross-correlations with output				
Consumption	0.87	0.84	0.92	0.81
Investment	0.91	0.88	0.94	0.86
Real exchange rate	-0.13	-0.09	0.41	-0.18
Net exports / GDP	-0.47	-0.44	-0.51	-0.49

Notes: Data moments from G7 countries, 1973Q1-2019Q4, HP-filtered. Baseline model includes nominal rigidities (Calvo pricing) and monetary policy rule. No Rigidities is a flex-price version. Financial Frictions adds Bernanke-Gertler-Gilchrist accelerator.

We make several simplifying assumptions to maintain tractability. First, we assume that agents have rational expectations and perfect information about the distribution of shocks. Second, we abstract from general equilibrium effects by treating prices as exogenous to individual agents. Third, we assume that the production technology exhibits constant returns to scale. We discuss the robustness of our theoretical predictions to relaxing each

of these assumptions in Appendix A, where we also present a more general version of the model.

The equilibrium of the model can be characterized by a system of first-order conditions and market-clearing conditions. In the interior solution, the marginal benefit of increasing the treatment variable equals the marginal cost, yielding an implicit equation for the optimal level of the treatment variable as a function of exogenous parameters and state variables. We log-linearize this equation around the steady state to obtain the estimating equation that motivates our empirical specification.

IV Solution Method

This paper investigates the causal relationship between the treatment variable and the outcome variable. Understanding this relationship is critical for both economic theory and policy. Despite a growing literature, credible identification of the causal effect remains challenging due to potential confounders and reverse causality. We contribute to this literature by proposing a new identification strategy based on an instrumental variables strategy. Our approach addresses the key threats to internal validity that have limited the conclusions of prior work.

We contribute to the literature on international macro by exploiting a novel source of variation in the treatment variable. Our identification strategy relies on an instrumental variables strategy, which allows us to isolate the causal effect of the treatment variable on the outcome variable from confounding factors that have plagued earlier studies. The identifying assumption underlying our approach is that the proposed instrument affects the outcome variable only through its effect on the treatment variable, conditional on observable characteristics. We provide several pieces of evidence in support of this assumption, including placebo tests, overidentification tests, and bounds analysis.

The existing literature on the relationship between the treatment variable and the out-

come variable has produced mixed results. Some studies find a positive and significant association, while others find no statistically significant effect. We argue that these discrepancies arise from differences in identification strategies and sample composition. Studies relying on cross-sectional variation tend to overstate the relationship due to omitted variable bias, while panel data approaches that rely on within-unit variation may attenuate the true effect by absorbing long-run changes in the treatment variable. Our approach navigates between these extremes by exploiting quasi-experimental variation.

A central question in international macro is whether changes in the treatment variable lead to changes in the outcome variable, or whether the observed correlation reflects reverse causality or omitted variable bias. We address this question using an instrumental variables strategy applied to data from administrative records. The institutional setting we study provides a compelling natural experiment because the relevant policy changes were driven by factors orthogonal to contemporaneous economic conditions. We document this independence using a battery of balance tests and falsification exercises.

Our main finding is that the treatment variable has a statistically significant and economically meaningful effect on the outcome variable. The magnitude of this effect is robust across a wide range of specifications, subsamples, and alternative measures of the key variables. A one standard deviation increase in the treatment variable is associated with a change of approximately 0.3 to 0.5 standard deviations in the outcome variable, depending on the specification. This magnitude is consistent with the predictions of the theoretical model developed in Section 2 and is comparable to estimates from related quasi-experimental studies.

The remainder of this paper proceeds as follows. Section 2 presents the theoretical framework that motivates our empirical analysis and generates testable predictions. Section 3 describes the institutional background and the natural experiment we exploit. Section 4 describes the data sources and sample construction. Section 5 outlines our empirical strategy, including the identification assumptions and potential threats to validity. Section

6 presents the main results. Section 7 discusses robustness checks and extensions. Section 8 concludes with a discussion of policy implications.

Our paper relates to several strands of the economics literature. First, we contribute to the extensive body of work on the determinants of the outcome variable, which has been a central topic in international macro since the seminal contributions of the 1990s and 2000s. Second, we contribute to the methodological literature on an instrumental variables strategy by demonstrating its applicability to a new empirical context. Third, our findings speak to an ongoing policy debate about the effectiveness of interventions targeting the treatment variable.

The economic significance of our findings warrants emphasis. Back-of-the-envelope calculations suggest that the estimated effect implies substantial welfare gains from policies that improve the treatment variable. Using the framework of Chetty (2009) for translating reduced-form estimates into welfare statements, we estimate that a policy intervention that increases the treatment variable by one standard deviation would generate welfare gains equivalent to approximately 2-4 percent of annual income. These calculations are necessarily approximate but suggest that the magnitudes involved are economically meaningful.

A distinctive feature of our analysis is the richness of the data. Unlike many previous studies that rely on aggregate proxies, our dataset contains individual-level information on both the outcome variable and the treatment variable, measured with high frequency over an extended time period. This granularity allows us to investigate heterogeneity in treatment effects across individuals, to construct precise event study estimates, and to conduct a range of placebo and falsification tests that would not be feasible with more aggregated data.

Before proceeding, we note several contributions relative to the closest existing work. While prior studies have documented a correlation between the treatment variable and the outcome variable in various contexts, our paper is among the first to provide credi-

bly causal estimates using an instrumental variables strategy. Moreover, our analysis of mechanisms goes beyond what has been possible in earlier studies, shedding light on the channels through which the treatment variable affects the outcome variable.

This paper investigates the causal relationship between the treatment variable and the outcome variable. Understanding this relationship is critical for both economic theory and policy. Despite a growing literature, credible identification of the causal effect remains challenging due to potential confounders and reverse causality. We contribute to this literature by proposing a new identification strategy based on an instrumental variables strategy. Our approach addresses the key threats to internal validity that have limited the conclusions of prior work.

We contribute to the literature on international macro by exploiting a novel source of variation in the treatment variable. Our identification strategy relies on an instrumental variables strategy, which allows us to isolate the causal effect of the treatment variable on the outcome variable from confounding factors that have plagued earlier studies. The identifying assumption underlying our approach is that the proposed instrument affects the outcome variable only through its effect on the treatment variable, conditional on observable characteristics. We provide several pieces of evidence in support of this assumption, including placebo tests, overidentification tests, and bounds analysis.

V Quantitative Results

Table **??** presents our main estimates. Column (1) reports the baseline specification without controls. Columns (2) through (4) progressively add demographic controls, geographic controls, and fixed effects. The point estimate remains remarkably stable across specifications, suggesting that selection on unobservables is unlikely to drive our results. Following the bounding approach of Oster (2019), we show that the degree of selection on unobservables required to explain away our estimate would need to be substantially larger than

the degree of selection on observables.

The estimated coefficient on the treatment variable is statistically significant at the one percent level in all specifications. The magnitude implies that a one standard deviation increase in the treatment variable is associated with a positive and significant of approximately 0.3 standard deviations in the outcome variable. To put this in perspective, this effect size is comparable to the impact of an additional year of schooling on earnings, one of the most well-established causal relationships in economics. The 95 percent confidence interval is sufficiently narrow to rule out effects smaller than one-tenth of a standard deviation.

We find no evidence that our results are driven by pre-existing trends or differential selection. The event study coefficients are small and statistically insignificant in all pre-treatment periods, lending support to our parallel trends assumption. Moreover, the treatment effect emerges sharply at the time of the intervention rather than gradually, which is consistent with a causal interpretation and inconsistent with pre-existing trends that happen to accelerate at the time of treatment.

The first-stage relationship between the proposed instrument and the treatment variable is strong, with an F-statistic well above conventional thresholds for weak instruments. The first-stage coefficient is precisely estimated and has the expected sign. Following the recommendations of Andrews, Stock, and Sun (2019), we also report the effective F-statistic and the tF critical values. Under all weak-instrument diagnostics considered, we can reject the null of a weak first stage at the 5 percent level.

Heterogeneity analysis reveals that the treatment effect is concentrated among individuals with below-median baseline levels of the outcome variable. This pattern is consistent with the theoretical prediction that the marginal return to the treatment variable is decreasing. We also find suggestive evidence of heterogeneity along demographic dimensions, although the subsample estimates are less precisely estimated due to smaller sample sizes. The difference in treatment effects across subgroups is statistically significant at the

10 percent level.

We complement the regression analysis with non-parametric evidence on the treatment effect. Kernel density estimates of the distribution of the outcome variable show a clear rightward shift for treated individuals relative to untreated individuals. Quantile treatment effects, reported in the appendix, are positive and statistically significant throughout the distribution, with somewhat larger effects in the lower tail. This pattern suggests that the treatment reduces inequality in the outcome variable by disproportionately benefiting those at the bottom of the distribution.

The magnitude of our estimates is informative about the relative importance of the treatment variable as a determinant of the outcome variable. A variance decomposition exercise reveals that variation in the treatment variable can account for approximately 15 to 25 percent of the cross-sectional variance in the outcome variable, depending on the specification. While this leaves substantial residual variation unexplained, it establishes the treatment variable as one of the most quantitatively important determinants identified in the literature.

We also examine the dynamics of the treatment effect by interacting the treatment variable with indicators for different time horizons. The effect is small and statistically insignificant in the first year after treatment, grows steadily over the subsequent three years, and stabilizes at a new long-run level by year four. This gradual emergence of the treatment effect is consistent with the theoretical model, which predicts that the adjustment to the new equilibrium occurs over multiple periods as agents update their behavior and as general equilibrium effects propagate through the economy.

Table **??** presents our main estimates. Column (1) reports the baseline specification without controls. Columns (2) through (4) progressively add demographic controls, geographic controls, and fixed effects. The point estimate remains remarkably stable across specifications, suggesting that selection on unobservables is unlikely to drive our results. Following the bounding approach of Oster (2019), we show that the degree of selection

on unobservables required to explain away our estimate would need to be substantially larger than the degree of selection on observables.

Table 3: Impulse Responses to a Home Productivity Shock (1% Innovation)

	Impact	1 Year	2 Years	5 Years
Panel A: Home country				
Output	0.71	0.58	0.43	0.14
Consumption	0.38	0.37	0.35	0.21
Investment	1.84	1.52	1.11	0.39
Real exchange rate (depreciation +)	-0.62	-0.44	-0.28	-0.07
Current account / GDP	0.41	0.29	0.18	0.03
Panel B: Foreign country (spillovers)				
Output	0.12	0.11	0.09	0.04
Consumption	0.18	0.17	0.14	0.07
Terms of trade (home TOT, deterioration +)	0.31	0.22	0.14	0.04

Notes: All responses are percentage deviations from steady state. Shock is a 1 standard deviation (0.71%) innovation to home total factor productivity. Monetary policy follows the estimated Taylor rule. Shaded regions are 90% confidence bands from 500 bootstrap replications.

The estimated coefficient on the treatment variable is statistically significant at the one percent level in all specifications. The magnitude implies that a one standard deviation increase in the treatment variable is associated with a positive and significant of approximately 0.3 standard deviations in the outcome variable. To put this in perspective, this effect size is comparable to the impact of an additional year of schooling on earnings, one of the most well-established causal relationships in economics. The 95 percent confidence interval is sufficiently narrow to rule out effects smaller than one-tenth of a standard deviation.

We find no evidence that our results are driven by pre-existing trends or differential selection. The event study coefficients are small and statistically insignificant in all pre-treatment periods, lending support to our parallel trends assumption. Moreover, the treatment effect emerges sharply at the time of the intervention rather than gradually, which is consistent with a causal interpretation and inconsistent with pre-existing trends that happen to accelerate at the time of treatment.

Table 4: Impulse Responses to a Home Monetary Policy Shock (25 bp Tightening)

	Impact	1 Year	2 Years	5 Years
Panel A: Home country				
Output	-0.22	-0.38	-0.29	-0.06
Inflation (annualized pp)	-0.09	-0.27	-0.31	-0.08
Nominal interest rate (pp)	0.25	0.19	0.11	0.02
Real exchange rate (appreciation +)	0.84	0.61	0.38	0.07
Panel B: Foreign country (spillovers)				
Output	-0.07	-0.11	-0.08	-0.02
Inflation (annualized pp)	-0.03	-0.09	-0.11	-0.03
Nominal interest rate (pp)	0.04	0.07	0.06	0.01

Notes: Responses are percentage deviations from steady state (except interest rates and inflation, which are percentage point deviations). The monetary shock is identified as an unexpected 25 bp increase in the policy rate.

The first-stage relationship between the proposed instrument and the treatment variable is strong, with an F-statistic well above conventional thresholds for weak instruments. The first-stage coefficient is precisely estimated and has the expected sign. Following the recommendations of Andrews, Stock, and Sun (2019), we also report the effective F-statistic and the tF critical values. Under all weak-instrument diagnostics considered, we can reject the null of a weak first stage at the 5 percent level.

A Impulse Responses to Technology Shocks

This paper investigates the causal relationship between the treatment variable and the outcome variable. Understanding this relationship is critical for both economic theory and policy. Despite a growing literature, credible identification of the causal effect remains challenging due to potential confounders and reverse causality. We contribute to this literature by proposing a new identification strategy based on an instrumental variables strategy. Our approach addresses the key threats to internal validity that have limited the conclusions of prior work.

We contribute to the literature on international macro by exploiting a novel source of variation in the treatment variable. Our identification strategy relies on an instrumental variables strategy, which allows us to isolate the causal effect of the treatment variable on the outcome variable from confounding factors that have plagued earlier studies. The identifying assumption underlying our approach is that the proposed instrument affects the outcome variable only through its effect on the treatment variable, conditional on observable characteristics. We provide several pieces of evidence in support of this assumption, including placebo tests, overidentification tests, and bounds analysis.

The existing literature on the relationship between the treatment variable and the outcome variable has produced mixed results. Some studies find a positive and significant association, while others find no statistically significant effect. We argue that these discrepancies arise from differences in identification strategies and sample composition. Studies relying on cross-sectional variation tend to overstate the relationship due to omitted variable bias, while panel data approaches that rely on within-unit variation may attenuate the true effect by absorbing long-run changes in the treatment variable. Our approach navigates between these extremes by exploiting quasi-experimental variation.

A central question in international macro is whether changes in the treatment variable lead to changes in the outcome variable, or whether the observed correlation reflects reverse causality or omitted variable bias. We address this question using an instrumental variables strategy applied to data from administrative records. The institutional setting we study provides a compelling natural experiment because the relevant policy changes were driven by factors orthogonal to contemporaneous economic conditions. We document this independence using a battery of balance tests and falsification exercises.

Our main finding is that the treatment variable has a statistically significant and economically meaningful effect on the outcome variable. The magnitude of this effect is robust across a wide range of specifications, subsamples, and alternative measures of the key variables. A one standard deviation increase in the treatment variable is associated

with a change of approximately 0.3 to 0.5 standard deviations in the outcome variable, depending on the specification. This magnitude is consistent with the predictions of the theoretical model developed in Section 2 and is comparable to estimates from related quasi-experimental studies.

The remainder of this paper proceeds as follows. Section 2 presents the theoretical framework that motivates our empirical analysis and generates testable predictions. Section 3 describes the institutional background and the natural experiment we exploit. Section 4 describes the data sources and sample construction. Section 5 outlines our empirical strategy, including the identification assumptions and potential threats to validity. Section 6 presents the main results. Section 7 discusses robustness checks and extensions. Section 8 concludes with a discussion of policy implications.

Our paper relates to several strands of the economics literature. First, we contribute to the extensive body of work on the determinants of the outcome variable, which has been a central topic in international macro since the seminal contributions of the 1990s and 2000s. Second, we contribute to the methodological literature on an instrumental variables strategy by demonstrating its applicability to a new empirical context. Third, our findings speak to an ongoing policy debate about the effectiveness of interventions targeting the treatment variable.

The economic significance of our findings warrants emphasis. Back-of-the-envelope calculations suggest that the estimated effect implies substantial welfare gains from policies that improve the treatment variable. Using the framework of Chetty (2009) for translating reduced-form estimates into welfare statements, we estimate that a policy intervention that increases the treatment variable by one standard deviation would generate welfare gains equivalent to approximately 2-4 percent of annual income. These calculations are necessarily approximate but suggest that the magnitudes involved are economically meaningful.

A distinctive feature of our analysis is the richness of the data. Unlike many previous

studies that rely on aggregate proxies, our dataset contains individual-level information on both the outcome variable and the treatment variable, measured with high frequency over an extended time period. This granularity allows us to investigate heterogeneity in treatment effects across individuals, to construct precise event study estimates, and to conduct a range of placebo and falsification tests that would not be feasible with more aggregated data.

Before proceeding, we note several contributions relative to the closest existing work. While prior studies have documented a correlation between the treatment variable and the outcome variable in various contexts, our paper is among the first to provide credibly causal estimates using an instrumental variables strategy. Moreover, our analysis of mechanisms goes beyond what has been possible in earlier studies, shedding light on the channels through which the treatment variable affects the outcome variable.

B Impulse Responses to Monetary Policy Shocks

This paper investigates the causal relationship between the treatment variable and the outcome variable. Understanding this relationship is critical for both economic theory and policy. Despite a growing literature, credible identification of the causal effect remains challenging due to potential confounders and reverse causality. We contribute to this literature by proposing a new identification strategy based on an instrumental variables strategy. Our approach addresses the key threats to internal validity that have limited the conclusions of prior work.

We contribute to the literature on international macro by exploiting a novel source of variation in the treatment variable. Our identification strategy relies on an instrumental variables strategy, which allows us to isolate the causal effect of the treatment variable on the outcome variable from confounding factors that have plagued earlier studies. The identifying assumption underlying our approach is that the proposed instrument affects the outcome variable only through its effect on the treatment variable, conditional on observ-

able characteristics. We provide several pieces of evidence in support of this assumption, including placebo tests, overidentification tests, and bounds analysis.

The existing literature on the relationship between the treatment variable and the outcome variable has produced mixed results. Some studies find a positive and significant association, while others find no statistically significant effect. We argue that these discrepancies arise from differences in identification strategies and sample composition. Studies relying on cross-sectional variation tend to overstate the relationship due to omitted variable bias, while panel data approaches that rely on within-unit variation may attenuate the true effect by absorbing long-run changes in the treatment variable. Our approach navigates between these extremes by exploiting quasi-experimental variation.

A central question in international macro is whether changes in the treatment variable lead to changes in the outcome variable, or whether the observed correlation reflects reverse causality or omitted variable bias. We address this question using an instrumental variables strategy applied to data from administrative records. The institutional setting we study provides a compelling natural experiment because the relevant policy changes were driven by factors orthogonal to contemporaneous economic conditions. We document this independence using a battery of balance tests and falsification exercises.

Our main finding is that the treatment variable has a statistically significant and economically meaningful effect on the outcome variable. The magnitude of this effect is robust across a wide range of specifications, subsamples, and alternative measures of the key variables. A one standard deviation increase in the treatment variable is associated with a change of approximately 0.3 to 0.5 standard deviations in the outcome variable, depending on the specification. This magnitude is consistent with the predictions of the theoretical model developed in Section 2 and is comparable to estimates from related quasi-experimental studies.

The remainder of this paper proceeds as follows. Section 2 presents the theoretical framework that motivates our empirical analysis and generates testable predictions. Sec-

tion 3 describes the institutional background and the natural experiment we exploit. Section 4 describes the data sources and sample construction. Section 5 outlines our empirical strategy, including the identification assumptions and potential threats to validity. Section 6 presents the main results. Section 7 discusses robustness checks and extensions. Section 8 concludes with a discussion of policy implications.

Our paper relates to several strands of the economics literature. First, we contribute to the extensive body of work on the determinants of the outcome variable, which has been a central topic in international macro since the seminal contributions of the 1990s and 2000s. Second, we contribute to the methodological literature on an instrumental variables strategy by demonstrating its applicability to a new empirical context. Third, our findings speak to an ongoing policy debate about the effectiveness of interventions targeting the treatment variable.

The economic significance of our findings warrants emphasis. Back-of-the-envelope calculations suggest that the estimated effect implies substantial welfare gains from policies that improve the treatment variable. Using the framework of Chetty (2009) for translating reduced-form estimates into welfare statements, we estimate that a policy intervention that increases the treatment variable by one standard deviation would generate welfare gains equivalent to approximately 2-4 percent of annual income. These calculations are necessarily approximate but suggest that the magnitudes involved are economically meaningful.

A distinctive feature of our analysis is the richness of the data. Unlike many previous studies that rely on aggregate proxies, our dataset contains individual-level information on both the outcome variable and the treatment variable, measured with high frequency over an extended time period. This granularity allows us to investigate heterogeneity in treatment effects across individuals, to construct precise event study estimates, and to conduct a range of placebo and falsification tests that would not be feasible with more aggregated data.

Before proceeding, we note several contributions relative to the closest existing work. While prior studies have documented a correlation between the treatment variable and the outcome variable in various contexts, our paper is among the first to provide credibly causal estimates using an instrumental variables strategy. Moreover, our analysis of mechanisms goes beyond what has been possible in earlier studies, shedding light on the channels through which the treatment variable affects the outcome variable.

C Variance Decomposition and Historical Analysis

This paper investigates the causal relationship between the treatment variable and the outcome variable. Understanding this relationship is critical for both economic theory and policy. Despite a growing literature, credible identification of the causal effect remains challenging due to potential confounders and reverse causality. We contribute to this literature by proposing a new identification strategy based on an instrumental variables strategy. Our approach addresses the key threats to internal validity that have limited the conclusions of prior work.

We contribute to the literature on international macro by exploiting a novel source of variation in the treatment variable. Our identification strategy relies on an instrumental variables strategy, which allows us to isolate the causal effect of the treatment variable on the outcome variable from confounding factors that have plagued earlier studies. The identifying assumption underlying our approach is that the proposed instrument affects the outcome variable only through its effect on the treatment variable, conditional on observable characteristics. We provide several pieces of evidence in support of this assumption, including placebo tests, overidentification tests, and bounds analysis.

The existing literature on the relationship between the treatment variable and the outcome variable has produced mixed results. Some studies find a positive and significant association, while others find no statistically significant effect. We argue that these discrepancies arise from differences in identification strategies and sample composition. Studies

relying on cross-sectional variation tend to overstate the relationship due to omitted variable bias, while panel data approaches that rely on within-unit variation may attenuate the true effect by absorbing long-run changes in the treatment variable. Our approach navigates between these extremes by exploiting quasi-experimental variation.

A central question in international macro is whether changes in the treatment variable lead to changes in the outcome variable, or whether the observed correlation reflects reverse causality or omitted variable bias. We address this question using an instrumental variables strategy applied to data from administrative records. The institutional setting we study provides a compelling natural experiment because the relevant policy changes were driven by factors orthogonal to contemporaneous economic conditions. We document this independence using a battery of balance tests and falsification exercises.

Our main finding is that the treatment variable has a statistically significant and economically meaningful effect on the outcome variable. The magnitude of this effect is robust across a wide range of specifications, subsamples, and alternative measures of the key variables. A one standard deviation increase in the treatment variable is associated with a change of approximately 0.3 to 0.5 standard deviations in the outcome variable, depending on the specification. This magnitude is consistent with the predictions of the theoretical model developed in Section 2 and is comparable to estimates from related quasi-experimental studies.

The remainder of this paper proceeds as follows. Section 2 presents the theoretical framework that motivates our empirical analysis and generates testable predictions. Section 3 describes the institutional background and the natural experiment we exploit. Section 4 describes the data sources and sample construction. Section 5 outlines our empirical strategy, including the identification assumptions and potential threats to validity. Section 6 presents the main results. Section 7 discusses robustness checks and extensions. Section 8 concludes with a discussion of policy implications.

Our paper relates to several strands of the economics literature. First, we contribute

to the extensive body of work on the determinants of the outcome variable, which has been a central topic in international macro since the seminal contributions of the 1990s and 2000s. Second, we contribute to the methodological literature on an instrumental variables strategy by demonstrating its applicability to a new empirical context. Third, our findings speak to an ongoing policy debate about the effectiveness of interventions targeting the treatment variable.

Table 5: Variance Decomposition: Share of Variance Explained by Each Shock (%)

	Home TFP	Foreign TFP	Home Mon.	Foreign Mon.	Cost-push
Panel A: Home country variables					
Output	62.4	8.3	14.2	4.1	11.0
Consumption	48.7	12.1	18.4	6.3	14.5
Inflation	11.3	3.2	31.4	9.8	44.3
Interest rate	24.1	5.8	41.7	12.3	16.1
Panel B: International variables					
Real exchange rate	21.4	19.8	27.3	25.6	5.9
Terms of trade	31.2	29.4	18.1	17.8	3.5
Current account / GDP	44.6	38.7	8.1	7.2	1.4

Notes: Variance decomposition at the business cycle frequency (6-32 quarters). Columns sum to 100 within rounding error. Home TFP = home total factor productivity shock; Home Mon. = home monetary policy shock.

Table 6: Historical Decomposition of Real Exchange Rate Fluctuations, 1980-2019 (Share of Variance, %)

	1980-1989	1990-1999	2000-2009	2010-2019
Productivity shocks	28.4	31.2	24.6	22.1
Monetary policy shocks	41.2	29.8	33.7	38.4
Demand shocks	18.6	24.3	21.4	19.8
Cost-push shocks	11.8	14.7	20.3	19.7

Notes: Rows sum to 100 within rounding error. Structural shocks identified via sign restrictions following Canova-De Nicolo (2002). Columns correspond to decade subperiods.

The economic significance of our findings warrants emphasis. Back-of-the-envelope calculations suggest that the estimated effect implies substantial welfare gains from policies that improve the treatment variable. Using the framework of Chetty (2009) for translat-

ing reduced-form estimates into welfare statements, we estimate that a policy intervention that increases the treatment variable by one standard deviation would generate welfare gains equivalent to approximately 2-4 percent of annual income. These calculations are necessarily approximate but suggest that the magnitudes involved are economically meaningful.

A distinctive feature of our analysis is the richness of the data. Unlike many previous studies that rely on aggregate proxies, our dataset contains individual-level information on both the outcome variable and the treatment variable, measured with high frequency over an extended time period. This granularity allows us to investigate heterogeneity in treatment effects across individuals, to construct precise event study estimates, and to conduct a range of placebo and falsification tests that would not be feasible with more aggregated data.

Before proceeding, we note several contributions relative to the closest existing work. While prior studies have documented a correlation between the treatment variable and the outcome variable in various contexts, our paper is among the first to provide credibly causal estimates using an instrumental variables strategy. Moreover, our analysis of mechanisms goes beyond what has been possible in earlier studies, shedding light on the channels through which the treatment variable affects the outcome variable.

VI Sensitivity Analysis

We conduct a battery of robustness checks to assess the sensitivity of our main findings. Table **??** reports results from alternative specifications that vary the set of controls, the sample definition, the functional form, and the estimation method. Across all specifications, the point estimate on the treatment variable remains within the 95 percent confidence interval of the baseline estimate, and the qualitative conclusions are unchanged.

Our results are robust to alternative definitions of the key variables. Using log transfor-

mations instead of levels, measuring the treatment variable at different points in time, and winsorizing outliers at the first and ninety-ninth percentiles all yield qualitatively similar estimates. We also consider alternative measures of the outcome variable from different data sources and find highly consistent results. The correlation between alternative measures exceeds 0.9, suggesting that measurement error is not a first-order concern.

Placebo tests using pre-determined outcomes and lagged dependent variables produce estimates that are small, statistically insignificant, and precisely estimated around zero. These null results provide evidence against the hypothesis that our findings are driven by pre-existing trends or by unobserved confounders that are correlated with the timing of the treatment. We also conduct a randomization inference exercise that permutes the treatment assignment across individuals and find that the true estimate lies in the extreme tail of the placebo distribution.

We also verify that our results are not sensitive to the choice of clustering level. Clustering standard errors at the state level, the county level, or using two-way clustering on state and year all yield similar inference. The point estimates are identical by construction; only the standard errors differ. Wild cluster bootstrap p-values, which are recommended when the number of clusters is small, confirm the statistical significance of our main result at conventional levels.

As an additional robustness check, we estimate the model using alternative econometric methods. Control function approaches, matching estimators, and regression discontinuity designs applied to subsamples near relevant cutoffs all yield estimates consistent with our baseline specification. The concordance of results across methods with different identifying assumptions strengthens the causal interpretation of our findings.

We address the possibility of spillover effects by examining whether the treatment of neighboring individuals affects outcomes for untreated individuals. Using the approach of Miguel and Kremer (2004), we estimate a model that allows for spatial spillovers and find that the spillover effects are small and statistically insignificant. This suggests that our

treatment effect estimates are not biased by violations of the stable unit treatment value assumption.

Finally, we conduct a leave-one-out analysis to assess the influence of individual observations on our results. Sequentially dropping each individuals from the sample and re-estimating the main specification reveals that no single observation has a disproportionate influence on the point estimate. The coefficient on the treatment variable varies by less than 10 percent across leave-one-out samples, and the statistical significance is maintained in all cases.

Table 7: Exchange Rate Disconnect: Correlations Between Exchange Rates and Macro Aggregates

	Data	Baseline	Incomplete Mkts	PCP Pricing
Corr(RER, rel. output)	-0.13	-0.09	-0.21	0.58
Corr(RER, rel. consumption)	0.08	0.11	0.71	0.64
Corr(NER, price level)	0.04	0.07	0.03	0.39
Backus-Smith correlation	0.08	0.14	0.68	0.61
RER volatility / relative output vol.	6.69	5.77	3.12	1.84
NER volatility / relative price vol.	7.34	6.98	6.41	1.72

Notes: Data: G7, 1973Q1-2019Q4. RER = real exchange rate, NER = nominal exchange rate, PCP = producer currency pricing. Backus-Smith correlation is corr(RER, C/C*). Baseline uses local currency pricing (LCP); PCP pricing uses producer currency.

We conduct a battery of robustness checks to assess the sensitivity of our main findings. Table **??** reports results from alternative specifications that vary the set of controls, the sample definition, the functional form, and the estimation method. Across all specifications, the point estimate on the treatment variable remains within the 95 percent confidence interval of the baseline estimate, and the qualitative conclusions are unchanged.

Our results are robust to alternative definitions of the key variables. Using log transformations instead of levels, measuring the treatment variable at different points in time, and winsorizing outliers at the first and ninety-ninth percentiles all yield qualitatively similar estimates. We also consider alternative measures of the outcome variable from different data sources and find highly consistent results. The correlation between alternative mea-

Table 8: Current Account Dynamics: Empirical Evidence and Model Fit

	(1)	(2)	(3)	(4)
Dep. var.: Current account / GDP (t+1)				
Current account / GDP (t)	0.71***	0.68***	0.66***	0.64***
	((0.04))	((0.04))	((0.05))	((0.05))
Relative output growth		-0.31***	-0.28***	-0.27***
		((0.08))	((0.08))	((0.08))
Real exchange rate change			0.04**	0.04**
			((0.02))	((0.02))
Terms of trade change				0.12***
				((0.04))
Country FE	No	Yes	Yes	Yes
Year FE	No	No	Yes	Yes
Observations	780	780	780	780
R^2	0.51	0.57	0.61	0.63

Notes: *** p<0.01, ** p<0.05, * p<0.1. Panel of 20 OECD countries, 1980-2019. Standard errors clustered by country.

Table 9: Terms of Trade and International Transmission

	(1)	(2)	(3)	(4)
Dep. var.: Log terms of trade				
Log relative productivity	-0.44***	-0.41***	-0.38***	-0.36***
	((0.07))	((0.07))	((0.08))	((0.08))
Log relative money supply		0.18***	0.16***	0.15***
		((0.05))	((0.05))	((0.05))
Trade balance / GDP			-0.21**	-0.19**
			((0.09))	((0.09))
Relative fiscal balance				0.07
				((0.06))
Country-pair FE	No	Yes	Yes	Yes
Year FE	No	No	Yes	Yes
Observations	1560	1560	1560	1560
R^2	0.22	0.41	0.47	0.48

Notes: *** p<0.01, ** p<0.05, * p<0.1. Sample: 40 country pairs (G7 cross-pairs), 1980-2019. Standard errors clustered by country pair.

sures exceeds 0.9, suggesting that measurement error is not a first-order concern.

Placebo tests using pre-determined outcomes and lagged dependent variables produce estimates that are small, statistically insignificant, and precisely estimated around zero. These null results provide evidence against the hypothesis that our findings are driven by pre-existing trends or by unobserved confounders that are correlated with the timing of the treatment. We also conduct a randomization inference exercise that permutes the treatment assignment across individuals and find that the true estimate lies in the extreme tail of the placebo distribution.

A Exchange Rate Disconnect

This paper investigates the causal relationship between the treatment variable and the outcome variable. Understanding this relationship is critical for both economic theory and policy. Despite a growing literature, credible identification of the causal effect remains challenging due to potential confounders and reverse causality. We contribute to this literature by proposing a new identification strategy based on an instrumental variables strategy. Our approach addresses the key threats to internal validity that have limited the conclusions of prior work.

We contribute to the literature on international macro by exploiting a novel source of variation in the treatment variable. Our identification strategy relies on an instrumental variables strategy, which allows us to isolate the causal effect of the treatment variable on the outcome variable from confounding factors that have plagued earlier studies. The identifying assumption underlying our approach is that the proposed instrument affects the outcome variable only through its effect on the treatment variable, conditional on observable characteristics. We provide several pieces of evidence in support of this assumption, including placebo tests, overidentification tests, and bounds analysis.

The existing literature on the relationship between the treatment variable and the outcome variable has produced mixed results. Some studies find a positive and significant

association, while others find no statistically significant effect. We argue that these discrepancies arise from differences in identification strategies and sample composition. Studies relying on cross-sectional variation tend to overstate the relationship due to omitted variable bias, while panel data approaches that rely on within-unit variation may attenuate the true effect by absorbing long-run changes in the treatment variable. Our approach navigates between these extremes by exploiting quasi-experimental variation.

A central question in international macro is whether changes in the treatment variable lead to changes in the outcome variable, or whether the observed correlation reflects reverse causality or omitted variable bias. We address this question using an instrumental variables strategy applied to data from administrative records. The institutional setting we study provides a compelling natural experiment because the relevant policy changes were driven by factors orthogonal to contemporaneous economic conditions. We document this independence using a battery of balance tests and falsification exercises.

Our main finding is that the treatment variable has a statistically significant and economically meaningful effect on the outcome variable. The magnitude of this effect is robust across a wide range of specifications, subsamples, and alternative measures of the key variables. A one standard deviation increase in the treatment variable is associated with a change of approximately 0.3 to 0.5 standard deviations in the outcome variable, depending on the specification. This magnitude is consistent with the predictions of the theoretical model developed in Section 2 and is comparable to estimates from related quasi-experimental studies.

The remainder of this paper proceeds as follows. Section 2 presents the theoretical framework that motivates our empirical analysis and generates testable predictions. Section 3 describes the institutional background and the natural experiment we exploit. Section 4 describes the data sources and sample construction. Section 5 outlines our empirical strategy, including the identification assumptions and potential threats to validity. Section 6 presents the main results. Section 7 discusses robustness checks and extensions. Section

8 concludes with a discussion of policy implications.

Our paper relates to several strands of the economics literature. First, we contribute to the extensive body of work on the determinants of the outcome variable, which has been a central topic in international macro since the seminal contributions of the 1990s and 2000s. Second, we contribute to the methodological literature on an instrumental variables strategy by demonstrating its applicability to a new empirical context. Third, our findings speak to an ongoing policy debate about the effectiveness of interventions targeting the treatment variable.

The economic significance of our findings warrants emphasis. Back-of-the-envelope calculations suggest that the estimated effect implies substantial welfare gains from policies that improve the treatment variable. Using the framework of Chetty (2009) for translating reduced-form estimates into welfare statements, we estimate that a policy intervention that increases the treatment variable by one standard deviation would generate welfare gains equivalent to approximately 2-4 percent of annual income. These calculations are necessarily approximate but suggest that the magnitudes involved are economically meaningful.

B Current Account and Terms of Trade Dynamics

This paper investigates the causal relationship between the treatment variable and the outcome variable. Understanding this relationship is critical for both economic theory and policy. Despite a growing literature, credible identification of the causal effect remains challenging due to potential confounders and reverse causality. We contribute to this literature by proposing a new identification strategy based on an instrumental variables strategy. Our approach addresses the key threats to internal validity that have limited the conclusions of prior work.

We contribute to the literature on international macro by exploiting a novel source of variation in the treatment variable. Our identification strategy relies on an instrumental

variables strategy, which allows us to isolate the causal effect of the treatment variable on the outcome variable from confounding factors that have plagued earlier studies. The identifying assumption underlying our approach is that the proposed instrument affects the outcome variable only through its effect on the treatment variable, conditional on observable characteristics. We provide several pieces of evidence in support of this assumption, including placebo tests, overidentification tests, and bounds analysis.

The existing literature on the relationship between the treatment variable and the outcome variable has produced mixed results. Some studies find a positive and significant association, while others find no statistically significant effect. We argue that these discrepancies arise from differences in identification strategies and sample composition. Studies relying on cross-sectional variation tend to overstate the relationship due to omitted variable bias, while panel data approaches that rely on within-unit variation may attenuate the true effect by absorbing long-run changes in the treatment variable. Our approach navigates between these extremes by exploiting quasi-experimental variation.

A central question in international macro is whether changes in the treatment variable lead to changes in the outcome variable, or whether the observed correlation reflects reverse causality or omitted variable bias. We address this question using an instrumental variables strategy applied to data from administrative records. The institutional setting we study provides a compelling natural experiment because the relevant policy changes were driven by factors orthogonal to contemporaneous economic conditions. We document this independence using a battery of balance tests and falsification exercises.

Our main finding is that the treatment variable has a statistically significant and economically meaningful effect on the outcome variable. The magnitude of this effect is robust across a wide range of specifications, subsamples, and alternative measures of the key variables. A one standard deviation increase in the treatment variable is associated with a change of approximately 0.3 to 0.5 standard deviations in the outcome variable, depending on the specification. This magnitude is consistent with the predictions of the

theoretical model developed in Section 2 and is comparable to estimates from related quasi-experimental studies.

The remainder of this paper proceeds as follows. Section 2 presents the theoretical framework that motivates our empirical analysis and generates testable predictions. Section 3 describes the institutional background and the natural experiment we exploit. Section 4 describes the data sources and sample construction. Section 5 outlines our empirical strategy, including the identification assumptions and potential threats to validity. Section 6 presents the main results. Section 7 discusses robustness checks and extensions. Section 8 concludes with a discussion of policy implications.

Our paper relates to several strands of the economics literature. First, we contribute to the extensive body of work on the determinants of the outcome variable, which has been a central topic in international macro since the seminal contributions of the 1990s and 2000s. Second, we contribute to the methodological literature on an instrumental variables strategy by demonstrating its applicability to a new empirical context. Third, our findings speak to an ongoing policy debate about the effectiveness of interventions targeting the treatment variable.

The economic significance of our findings warrants emphasis. Back-of-the-envelope calculations suggest that the estimated effect implies substantial welfare gains from policies that improve the treatment variable. Using the framework of Chetty (2009) for translating reduced-form estimates into welfare statements, we estimate that a policy intervention that increases the treatment variable by one standard deviation would generate welfare gains equivalent to approximately 2-4 percent of annual income. These calculations are necessarily approximate but suggest that the magnitudes involved are economically meaningful.

VII Conclusion

This paper provides new evidence on the causal effect of the treatment variable on the outcome variable. Using an instrumental variables strategy applied to administrative records, we find a positive and significant and statistically significant effect that is robust across a wide range of specifications. The magnitude of the effect is economically meaningful and has clear implications for both theory and policy.

Our findings have several implications for economic theory. First, the results support models in which the treatment variable plays a central role in determining the outcome variable, as opposed to alternative theories that emphasize other factors. Second, the pattern of heterogeneity in treatment effects is consistent with the predictions of the theoretical framework, providing a rare example of a model that is both internally consistent and empirically validated. Third, the dynamics of the treatment effect shed light on the speed of adjustment to economic shocks, a parameter that is notoriously difficult to estimate.

From a policy perspective, the estimates suggest that interventions targeting the treatment variable could have meaningful effects on the outcome variable. However, several caveats are in order. First, our estimates are local to the population and context studied, and external validity to other settings should not be assumed without additional evidence. Second, our identification strategy relies on assumptions that, while supported by the evidence, cannot be directly tested. Third, the general equilibrium effects of large-scale policy interventions may differ from the partial equilibrium effects captured by our estimates.

Future work should investigate the mechanisms through which the treatment variable affects the outcome variable in greater detail. Our results identify a reduced-form relationship but do not fully pin down the structural channels at work. Complementary evidence from randomized controlled trials, structural estimation, or natural experiments in other contexts would strengthen the conclusions drawn here. Additionally, understanding the long-run persistence of the effects documented in this paper is an important question for future research.

In conclusion, this paper demonstrates that the treatment variable is a quantitatively important determinant of the outcome variable. The combination of a compelling identification strategy, rich data, and robust results across specifications provides a strong foundation for the causal claims made here. We hope that these findings will stimulate further research on the economic consequences of variation in the treatment variable and will inform the design of policies aimed at improving the outcome variable.

This paper provides new evidence on the causal effect of the treatment variable on the outcome variable. Using an instrumental variables strategy applied to administrative records, we find a positive and significant and statistically significant effect that is robust across a wide range of specifications. The magnitude of the effect is economically meaningful and has clear implications for both theory and policy.

Our findings have several implications for economic theory. First, the results support models in which the treatment variable plays a central role in determining the outcome variable, as opposed to alternative theories that emphasize other factors. Second, the pattern of heterogeneity in treatment effects is consistent with the predictions of the theoretical framework, providing a rare example of a model that is both internally consistent and empirically validated. Third, the dynamics of the treatment effect shed light on the speed of adjustment to economic shocks, a parameter that is notoriously difficult to estimate.

From a policy perspective, the estimates suggest that interventions targeting the treatment variable could have meaningful effects on the outcome variable. However, several caveats are in order. First, our estimates are local to the population and context studied, and external validity to other settings should not be assumed without additional evidence. Second, our identification strategy relies on assumptions that, while supported by the evidence, cannot be directly tested. Third, the general equilibrium effects of large-scale policy interventions may differ from the partial equilibrium effects captured by our estimates.

Future work should investigate the mechanisms through which the treatment variable affects the outcome variable in greater detail. Our results identify a reduced-form relation-

ship but do not fully pin down the structural channels at work. Complementary evidence from randomized controlled trials, structural estimation, or natural experiments in other contexts would strengthen the conclusions drawn here. Additionally, understanding the long-run persistence of the effects documented in this paper is an important question for future research.

In conclusion, this paper demonstrates that the treatment variable is a quantitatively important determinant of the outcome variable. The combination of a compelling identification strategy, rich data, and robust results across specifications provides a strong foundation for the causal claims made here. We hope that these findings will stimulate further research on the economic consequences of variation in the treatment variable and will inform the design of policies aimed at improving the outcome variable.

VIII Appendix A: Model Derivation

Consider an economy populated by a continuum of agents indexed by $i \in [0, 1]$. Each agent chooses the outcome variable to maximize lifetime utility subject to a budget constraint. The key source of heterogeneity across agents is their exposure to the treatment variable. We solve for the equilibrium of this economy and derive comparative statics with respect to the key parameters. The model generates a clear testable prediction: an exogenous increase in the treatment variable should lead to a positive and significant change in the outcome variable, with the magnitude depending on structural parameters that we estimate in the empirical analysis.

Let Y_i denote the outcome of interest for unit i. We model Y_i as a function of treatment status D_i, observed covariates X_i, and unobserved heterogeneity ε_i. Under standard regularity conditions, the conditional expectation function takes the form presented in Equation **??**. The parameter of interest is the coefficient on treatment status, which captures the causal effect of the treatment variable on the outcome variable holding constant ob-

served and unobserved confounders. Identification of this parameter requires an exclusion restriction, which we discuss in detail below.

The theoretical framework builds on the canonical model of field. Agents face a discrete choice between J alternatives, each yielding utility $U_{ij} = V_{ij} + \epsilon_{ij}$, where V_{ij} is the deterministic component and ϵ_{ij} captures idiosyncratic preferences. Under standard distributional assumptions on the error term, the choice probabilities take a tractable closed form. The model can be estimated using maximum likelihood or method of moments, and the resulting structural parameters can be used to simulate counterfactual policy scenarios.

We derive the key prediction of the model through a simple comparative statics exercise. An exogenous increase in the treatment variable shifts the equilibrium allocation of the outcome variable through two channels: a direct effect operating through prices, and an indirect effect operating through the reallocation of factors across sectors. The total effect is ambiguous in general, but under the empirically relevant parameter values, we show that the direct effect dominates. Proposition 1 in Appendix A formalizes this result and establishes conditions under which the sign of the total effect is unambiguous.

Under the assumption that the proposed instrument satisfies the exclusion restriction, the model implies that the reduced-form relationship between the proposed instrument and the outcome variable is proportional to the structural parameter of interest. This motivates our instrumental variables strategy described in Section **??**. A key advantage of the IV approach is that it does not require functional form assumptions on the relationship between the treatment variable and the unobserved confounders. The cost is that we identify a local average treatment effect rather than the population average treatment effect.

The model delivers additional testable implications beyond the main prediction. First, the treatment effect should be larger for individuals with lower baseline levels of the outcome variable, reflecting diminishing marginal returns. Second, the effect should be persistent over time rather than transitory, because the mechanism operates through permanent

changes in the level of the treatment variable. Third, the effect should be concentrated in sectors with greater exposure to the underlying channel. We test each of these auxiliary predictions in Section 6.

IX Appendix B: Log-Linearization

This appendix provides additional results referenced in the main text. We organize the supplementary material into three parts. Appendix A presents proofs of the theoretical propositions stated in Section **??**. Appendix B reports additional empirical results, including alternative specifications and sample definitions. Appendix C describes the construction of key variables in detail.

We present detailed derivations of the key theoretical results stated in Section **??**. The proofs proceed under standard regularity conditions that are verified in the data. We state these conditions formally as Assumptions A1 through A4 below, and provide discussion of their economic content and empirical plausibility. Throughout, we maintain the notation introduced in the main text.

This section reports results from additional robustness exercises not included in the main text due to space constraints. All estimates are qualitatively similar to the baseline results reported in Table **??**. We organize these results by the type of robustness check: alternative control variables, alternative sample restrictions, alternative functional forms, alternative estimation methods, and alternative definitions of key variables.

We provide here a detailed description of the data sources and variable construction procedures. For each variable, we report the original source, the years available, the unit of observation, and any transformations applied. We also report summary statistics for the full set of variables used in the analysis, including those that appear only in robustness checks or auxiliary regressions.

The following tables report the complete set of coefficient estimates for all control

variables included in our main specifications. In the interest of space, the main text reports only the coefficients on the key independent variables. The full results presented here confirm that the control variables enter with expected signs and magnitudes, and that their inclusion does not materially affect the coefficients of interest.

This appendix provides additional results referenced in the main text. We organize the supplementary material into three parts. Appendix A presents proofs of the theoretical propositions stated in Section **??**. Appendix B reports additional empirical results, including alternative specifications and sample definitions. Appendix C describes the construction of key variables in detail.

X Appendix C: Additional Results

Table **??** presents our main estimates. Column (1) reports the baseline specification without controls. Columns (2) through (4) progressively add demographic controls, geographic controls, and fixed effects. The point estimate remains remarkably stable across specifications, suggesting that selection on unobservables is unlikely to drive our results. Following the bounding approach of Oster (2019), we show that the degree of selection on unobservables required to explain away our estimate would need to be substantially larger than the degree of selection on observables.

The estimated coefficient on the treatment variable is statistically significant at the one percent level in all specifications. The magnitude implies that a one standard deviation increase in the treatment variable is associated with a positive and significant of approximately 0.3 standard deviations in the outcome variable. To put this in perspective, this effect size is comparable to the impact of an additional year of schooling on earnings, one of the most well-established causal relationships in economics. The 95 percent confidence interval is sufficiently narrow to rule out effects smaller than one-tenth of a standard deviation.

We find no evidence that our results are driven by pre-existing trends or differential selection. The event study coefficients are small and statistically insignificant in all pre-treatment periods, lending support to our parallel trends assumption. Moreover, the treatment effect emerges sharply at the time of the intervention rather than gradually, which is consistent with a causal interpretation and inconsistent with pre-existing trends that happen to accelerate at the time of treatment.

The first-stage relationship between the proposed instrument and the treatment variable is strong, with an F-statistic well above conventional thresholds for weak instruments. The first-stage coefficient is precisely estimated and has the expected sign. Following the recommendations of Andrews, Stock, and Sun (2019), we also report the effective F-statistic and the tF critical values. Under all weak-instrument diagnostics considered, we can reject the null of a weak first stage at the 5 percent level.

Heterogeneity analysis reveals that the treatment effect is concentrated among individuals with below-median baseline levels of the outcome variable. This pattern is consistent with the theoretical prediction that the marginal return to the treatment variable is decreasing. We also find suggestive evidence of heterogeneity along demographic dimensions, although the subsample estimates are less precisely estimated due to smaller sample sizes. The difference in treatment effects across subgroups is statistically significant at the 10 percent level.

Table 10: Sensitivity Analysis: Key Model Moments Under Alternative Parameterizations

	Baseline	Low sigma	High theta	Low xi	Open economy
RER volatility	9.87	7.41	8.12	11.34	10.28
RER-output corr.	-0.09	-0.07	-0.14	-0.11	-0.08
Backus-Smith corr.	0.14	0.42	0.09	0.08	0.21
CA volatility	1.76	1.54	2.11	1.83	2.34
TOT volatility	3.08	2.84	2.41	3.31	3.67

Notes: Baseline: sigma=2, theta=1.5, xi=0.75, alpha=0.72. Low sigma: sigma=1. High theta: theta=3. Low xi: xi=0.50 (more price flexibility). Open economy: alpha=0.50.

A.1 Household Lagrangian and First-Order Conditions

The representative household in the home country maximizes:

$$E_0 \sum_{t=0}^{\infty} \beta^t \left[\frac{C_t^{1-\sigma}}{1-\sigma} - \frac{L_t^{1+\phi}}{1+\phi} \right] \tag{11}$$

subject to the budget constraint (Equation **??**). The Lagrangian is:

$$\begin{aligned} \mathcal{L} = E_0 \sum_{t=0}^{\infty} \beta^t \Bigg[& \frac{C_t^{1-\sigma}}{1-\sigma} - \frac{L_t^{1+\phi}}{1+\phi} \\ & + \lambda_t \Big((1+i_{t-1})B_{t-1} + S_t(1+i_{t-1}^*)B_{t-1}^* + W_t L_t + T_t - P_t C_t - B_t - S_t B_t^* \Big) \Bigg] \end{aligned} \tag{12}$$

First-order conditions with respect to C_t, B_t, B_t^*, and L_t:

$$C_t^{-\sigma} = \lambda_t P_t \tag{13}$$

$$\lambda_t = \beta(1+i_t)E_t\lambda_{t+1} \tag{14}$$

$$\lambda_t S_t = \beta(1+i_t^*)E_t[\lambda_{t+1}S_{t+1}] \tag{15}$$

$$\frac{L_t^{\phi}}{W_t/P_t} = \lambda_t P_t \implies L_t^{\phi} = C_t^{-\sigma}\frac{W_t}{P_t} \tag{16}$$

Combining (**??**) and (**??**) yields the standard Euler equation (**??**). Dividing (**??**) by (**??**) gives the UIP condition (**??**) up to a Jensen's inequality correction term.

A.2 Log-Linearization Around Steady State

Let lower-case variables denote log-deviations from steady state: $\hat{x}_t \equiv \ln(X_t/\bar{X})$. The log-linearized Euler equation is:

$$\hat{c}_t = E_t\hat{c}_{t+1} - \frac{1}{\sigma}(i_t - E_t\pi_{t+1} - \bar{r}) \tag{17}$$

where $\bar{r} = 1/\beta - 1$ is the steady-state real interest rate. Log-linearizing the CES aggregator:

$$\hat{c}_t = (1-\alpha)\hat{c}_{H,t} + \alpha\hat{c}_{F,t} \tag{18}$$

and the demand system gives:

$$\hat{c}_{H,t} = -\theta(\hat{p}_{H,t} - \hat{p}_t) + \hat{c}_t \tag{19}$$

$$\hat{c}_{F,t} = -\theta(\hat{p}_{F,t} - \hat{p}_t) + \hat{c}_t \tag{20}$$

A.3 Blanchard-Kahn Conditions for Determinacy

The model can be written in state-space form:

$$\begin{pmatrix} \hat{\mathbf{x}}_{t+1} \\ E_t\hat{\mathbf{y}}_{t+1} \end{pmatrix} = \mathbf{A} \begin{pmatrix} \hat{\mathbf{x}}_t \\ \hat{\mathbf{y}}_t \end{pmatrix} + \mathbf{B}\hat{\varepsilon}_{t+1} \tag{21}$$

where $\hat{\mathbf{x}}_t$ is the vector of predetermined state variables and $\hat{\mathbf{y}}_t$ is the vector of jump variables. The Blanchard-Kahn conditions for a unique stable equilibrium require that the number of eigenvalues of $\mathbf{A}$ outside the unit circle equals the number of jump variables. Under the estimated Taylor rule parameters ($\phi_\pi = 1.5 > 1$), the model satisfies the Taylor principle and determinacy obtains.

A.4 Solution via Undetermined Coefficients

Conjecture that the solution takes the form $\hat{\mathbf{y}}_t = \mathbf{M}\hat{\mathbf{x}}_t$. Substituting into the system and matching coefficients:

$$\mathbf{M} = \mathbf{A}_{22}^{-1}(\mathbf{M} - \mathbf{A}_{21} - \mathbf{A}_{22}\mathbf{M}\mathbf{A}_{11} - \mathbf{M}\mathbf{A}_{11}) \tag{22}$$

This matrix equation can be solved via the QZ (generalized Schur) decomposition of $\mathbf{A}$, implemented in the Klein (2000) algorithm. The solution matrices $\mathbf{M}$ and the impulse response functions reported in Tables **??** and **??** are computed using this method.

A.5 Double Integral Representation of Exchange Rate Dynamics

The continuous-time real exchange rate satisfies:

$$\dot{q}_t = \iint_{\mathcal{F}_t \times \mathcal{F}_t^*} \frac{\partial^2 q}{\partial a \partial a^*} \, da \, da^* + \ddot{q}_t^{\text{residual}}, \tag{23}$$

where the double integral captures the second-order interaction between home and foreign productivity shocks, and the $\mathcal{F}_t$-measurable filtration defines the information set at time t.

A.6 Calligraphic Likelihood for Bayesian Estimation

The posterior density of the structural parameters $\boldsymbol{\theta}$ conditional on the observed data $\mathbf{Y}$ satisfies:

$$p(\boldsymbol{\theta} \mid \mathbf{Y}) \propto \mathcal{L}(\mathbf{Y} \mid \boldsymbol{\theta}) \cdot p(\boldsymbol{\theta}), \quad \mathcal{L}(\mathbf{Y} \mid \boldsymbol{\theta}) = \prod_{t=1}^{T} \mathcal{F}\Big(\dot{Y}_t \mid \dot{Y}_{t-1}, \boldsymbol{\theta}\Big). \tag{24}$$

We complement the regression analysis with non-parametric evidence on the treatment effect. Kernel density estimates of the distribution of the outcome variable show a clear rightward shift for treated individuals relative to untreated individuals. Quantile treatment effects, reported in the appendix, are positive and statistically significant throughout the distribution, with somewhat larger effects in the lower tail. This pattern suggests that the treatment reduces inequality in the outcome variable by disproportionately benefiting those at the bottom of the distribution.

References

Obstfeld, M. and K. Rogoff (1996). *Foundations of International Macroeconomics*. MIT Press.

Backus, D., P. Kehoe, and F. Kydland (1992). "International Real Business Cycles." *Journal of Political Economy*, 100(4), 745–775.

Gali, J. and M. Gertler (1999). "Inflation Dynamics: A Structural Econometric Analysis." *Journal of Monetary Economics*, 44(2), 195–222.

Clarida, R., J. Gali, and M. Gertler (1999). "The Science of Monetary Policy: A New Keynesian Perspective." *Journal of Economic Literature*, 37(4), 1661–1707.

Coeurdacier, N. and H. Rey (2013). "Home Bias in Open Economy Financial Macroeconomics." *Journal of Economic Literature*, 51(1), 63–115.

Engel, C. (1999). "Accounting for U.S. Real Exchange Rate Changes." *Journal of Political Economy*, 107(3), 507–538.

Betts, C. and M. Devereux (2000). "Exchange Rate Dynamics in a Model of Pricing-to-Market." *Journal of International Economics*, 50(1), 215–244.

Klein, P. (2000). "Using the Generalized Schur Form to Solve a Multivariate Linear Rational Expectations Model." *Journal of Economic Dynamics and Control*, 24(10), 1405–1423.